

Robust Trajectory Tracking Control for Underactuated Autonomous Underwater Vehicles

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Abstract—Motion control of underwater robotic vehicles is a demanding task with great challenges imposed by external disturbances, model uncertainties and constraints of the operating workspace. Thus, robust motion control is still an open issue for the underwater robotics community. In that sense, this paper addresses the tracking control problem of 3D trajectories for underactuated underwater robotic vehicles operating in a constrained workspace including obstacles. In particular, a robust Nonlinear Model Predictive Control (NMPC) scheme is presented for the case of underactuated Autonomous Underwater Vehicles (AUVs) (i.e., vehicles actuated only in surge, heave and yaw). The purpose of the controller is to steer the underactuated AUV to a desired trajectory with guaranteed input and state constraints inside a partially known and dynamic environment where the knowledge of the operating workspace is constantly updated on-line via the vehicle's on-board sensors. In particular, by considering a ball which covers the volume of the system, obstacle avoidance with any of the detected obstacles is guaranteed, despite the model dynamic uncertainties and the presence of external disturbances representing ocean currents and waves. The proposed feedback control law consists of two parts: an online law which is the outcome of a Finite Horizon Optimal Control Problem (FHOCP) solved for the nominal dynamics; and a state feedback law which is tuned off-line and guarantees that the real trajectories remain bounded in a hyper-tube centered along the nominal trajectories for all times. Finally, a simulation study verifies the performance and efficiency of the proposed approach.

I. INTRODUCTION

During the last decades, considerable progress has been made in the field of unmanned marine vehicles, with a significant number of results in a variety of marine activities [1]. Applications such as ocean forecasting, ecosystem monitoring, underwater inspection of oil/gas pipelines and risers are indicative examples of applications that require the underwater robots to work under various constraints (e.g., communication [2] energy [3] and sensing limitations) and increased level of autonomy in order to educing efficiently the various underwater resources [4] .

A typical marine control problem is trajectory tracking which aims to steer the Autonomous Underwater Vehicle on a reference trajectory[5]. Classical approaches such as local linearization and input-output decoupling have been used in the past to design motion controllers for underwater vehicles [6]. Nevertheless, the aforementioned methods yielded poor closed-loop performance and the results were local, around

only certain selected operating points. Output feedback linearization [7], [8] is an alternative approach which however is not always possible. Moreover, based on a combined approach involving Lyapunov theory and backstepping, various model-based non-linear controllers have been proposed in the literature [9]–[13]. More specifically, in [11], the authors extend their previous works in [9], [10], by proposing a nonlinear adaptive control law that drives an underwater vehicle along a sequence of desired way points. In the aforementioned control law, employing the backstepping technique, a kinematic controller law is extended to the dynamical model. In [12], the authors proposed a tracking controller by employing the integral backstepping technique. However, the effect of the second hydrodynamic damping force was ignored.

Dynamic model uncertainties of underwater robotic vehicles have been mainly compensated by employing adaptive control techniques [14], [15]. For instance, in [16], the authors proposed an adaptive control approach for an underwater robotic vehicle in presence of model uncertainties. However, owing to the sensitivity of the aforementioned controllers on unknown parameters, their capability in a real time experiment remains questionable. Moreover, based on switching control strategies and backstepping techniques, a hybrid parameter adaptation law was presented in [17], [18], where the effects of external disturbances and un-modeled dynamics were not considered. In addition, for all of the aforementioned motion control strategies, it is not always feasible or straightforward to incorporate input (generalized body forces/torques or thrust) and state (3D obstacles, velocities) constraints into the vehicle's closed-loop motion [19]. In that sense, the motion control problem of underwater robots continues to pose considerable challenges to system designers, especially in view of the high-demanding missions envisioned by the marine industry (e.g., ship hull inspection, surveillance of oil platforms, cable installation and tracking, etc.).

On the other hand, the reference/desired trajectory for the underwater robot is usually the result of some path planning techniques [20], [21]. The majority of planning techniques are based on off-line optimization schemes, which consider static or quasi-static operational environments. Their output is often a set of way-points or trajectories, which are optimal with respect to energy consumption, while satisfying certain environmental constraints (i.e., known obstacles). However, in real-time missions, the vehicle operates in a partially known and dynamic environment where the knowledge of the operating workspace is constantly updated on-line via the

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用于欠驱动自主水下航行器的鲁棒轨迹跟踪控制

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Abstract——水下机器人车辆的运动控制是一项具有挑战性的任务, 外部扰动、模型不确定性以及操作空间的约束都对其提出了很高的要求。因此, 鲁棒运动控制仍然是水下机器人社区的一个未解决问题。在这方面, 本文讨论了在包含障碍物的受限工作空间中运行的欠驱动水下机器人车辆的三维轨迹跟踪控制问题。具体来说, 对于欠驱动自主水下航行器 (AUV) (即仅在纵向、升降和偏航方向受控的车辆) 的情况, 本文提出了一种鲁棒非线性模型预测控制 (NMPC) 方案。该控制器的目的是在部分已知且动态变化的环境中, 使欠驱动AUV沿期望轨迹运动, 同时保证输入和状态约束, 其中操作空间的信息通过车辆的船载传感器不断在线更新。具体而言, 通过考虑

仅针对某些特定的选定运行点。输出反馈线性化 [7], [8] 是另一种方法, 但这种方法并非总是可行的。此外, 基于结合李雅普诺夫理论和反步法的方法, 文献中提出了各种基于模型的非线性控制器 [9]–[13]。更具体地说, 在 [11] 中, 作者在 [9]、[10] 的前期工作基础上, 提出了一种非线性自适应控制律, 使水下航行器沿一系列期望航点运动。在上述控制律中, 采用反步法技术, 将运动学控制律扩展到动力学模型中。在 [12] 中, 作者通过采用积分反步技术提出了一种跟踪控制器。然而, 第二个水动力阻尼力的影响被忽略了。

水下机器人车辆的动态模型不确定性主要通过采用自适应控制技术来补偿 [14], [15]。例如, 在 [16] 中, 作者提出了一种在存在模型不确定性情况下针对水下机器人车辆的自适应控制方法。然而, 由于上述控制器对未知参数的敏感性, 其在实时实验中的能力仍然存疑。此外, 基于切换控制策略和反步控制技术, 在 [17], [18] 中提出了一种混合参数自适应律, 但未考虑外部扰动和未建模动态的影响。此外, 对于所有上述运动控制策略, 将输入 (广义机体力/力矩或推力) 和状态 (3D障碍物、速度) 约束纳入车辆闭环运动中, 并不总是可行或简单 [19]。从这个意义上讲, 水下……

一、引言

在过去的几十年里, 无人海洋车辆领域取得了相当大的进展, 并在各种海洋活动中产生了大量成果 [1]。诸如海洋预报、生态系统监测、油气管道和立管的水下检查等应用, 是需要水下机器人在各种约束条件下工作 (例如, 通信 [2]、能源 [3] 和传感限制) 并提高自主水平以有效利用各种水下资源 [4] 的典型例子。

一个典型的海洋控制问题是轨迹跟踪, 其目标是引导自主水下航行器沿参考轨迹行驶 [5]。过去曾使用诸如局部线性化和输入输出解耦的经典方法为水下航行器设计运动控制器 [6]。然而, 上述方法产生了较差的闭环性能, 且结果是局部的, 围绕着

另一方面, 水下机器人的参考/期望轨迹通常是某些路径规划技术的结果 [20], [21]。大多数规划技术都是基于离线优化方案, 这些方案考虑的是静态或准静态的操作环境。它们的输出通常是一组航点或轨迹, 这些轨迹在满足特定环境约束 (即已知障碍物) 的前提下, 在能耗方面是最优的。然而, 在实时任务中, 车辆在部分已知和动态的环境中运行, 并且操作工作空间的知识会通过实时在线更新不断更新

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vehicle's on-board sensors (e.g., multi-beam imaging sonars, Onboard vision system, on-line ocean current estimators). In these cases, the underwater vehicle has to re-calculate its path on-line according to possible environmental changes (i.e., new obstacles, other vehicles or humans operating in proximity etc.). More details regarding the open problems on underwater robotics path planning, can be found in [22] and [23].

Motivated by the aforementioned considerations, this work presents a robust trajectory tracking control scheme for underactuated Autonomous Underwater Vehicles (AUVs) operating in a constrained workspace including obstacles. In particular, a robust Nonlinear Model Predictive Control (NMPC) scheme is presented for the underactuated AUVs (i.e., vehicles actuated only in surge, heave and yaw). Various constraints such as: sparse obstacles, workspace boundaries, control input saturation are considered during the control design. The purpose of the controller is to steer the underactuated AUV on a desired trajectory inside a constrained and dynamic workspace. Since the knowledge of the operating workspace is constantly updated online via the vehicle's on-board sensors, the robot re-calculates its path online if the updated environmental changes (i.e., new detected obstacles) are in conflict with the reference trajectory. In particular, by considering a ball which covers the volume of the system, obstacle avoidance with any of the detected obstacles is guaranteed, despite the model dynamic uncertainties and the presence of external disturbances representing ocean currents and waves. The proposed feedback control law consists of two parts: an online law which is the outcome of a Finite Horizon Optimal Control Problem (FHOC) solved for the nominal dynamics; and a state feedback law which is tuned off-line and guarantees that the real trajectories remain bounded in a hyper-tube centered along the nominal trajectories for all times. The volume of the hyper-tube depends on the upper bound of the disturbances as well as bounds of derivatives of the dynamics. The closed-loop system has analytically guaranteed stability and convergence properties. Finally, a simulation study verifies the performance and efficiency of the proposed approach.

II. NOTATION AND BACKGROUND

In this work, the vectors are denoted with lower bold letters whereas the matrices by capital bold letters. Define by $\mathbb{N}$ and $\mathbb{R}$ the sets of positive integers and real numbers, respectively. Given a set $\mathcal{S}$, denote by $|\mathcal{S}|$ and $\mathcal{S}^n := \mathcal{S} \times \dots \times \mathcal{S}$ its cardinality and its n -fold Cartesian product. Given a vector $\mathbf{z} \in \mathbb{R}^n$ define by:

$$\|\mathbf{z}\|_2 := \sqrt{\mathbf{z}^\top \mathbf{z}}, \quad \|\mathbf{z}\|_\infty := \max_{i=1,\dots,n} |z_i|, \quad \|\mathbf{z}\|_P := \sqrt{\mathbf{z}^\top \mathbf{P} \mathbf{z}},$$

its Euclidean, infinite and $\mathbf{P}$ -weighted norm, respectively, with $\mathbf{P} \geq 0$. The notation $\lambda_{\min}(\mathbf{P})$ stands for the minimum absolute value of the real part of the eigenvalues of $\mathbf{P} \in \mathbb{R}^{n \times n}$; $\mathbf{0}_{m \times n} \in \mathbb{R}^{m \times n}$ and $\mathbf{I}_n \in \mathbb{R}^{n \times n}$ stand for the $m \times n$ matrix with all entries zeros and the identity matrix, respectively. The notation $\text{diag}\{P_1, \dots, P_n\}$ stands for the

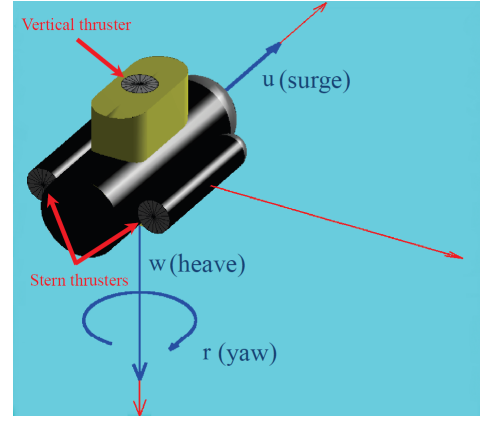


Fig. 1: The underactuated underwater vehicle. Blue color indicates the actuated degrees of freedom.

block diagonal matrix with the matrices $P_1, \dots, P_n$ in the main diagonal;

$$\mathcal{B}(\mathbf{c}, r) := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{c}\|_2 \leq r\},$$

stands for a ball in $\mathbb{R}^n$ with center and radius $\mathbf{c} \in \mathbb{R}^n$, $r > 0$, respectively. Given sets $\mathcal{S}_1, \mathcal{Z} \subseteq \mathbb{R}^n$, $\mathcal{S}_2 \subseteq \mathbb{R}^m$ and matrix $\mathbf{P} \in \mathbb{R}^{n \times m}$, the *Minkowski addition*, the *Pontryagin difference* and the *matrix-set multiplication* are respectively defined by:

$$\mathcal{S}_1 \oplus \mathcal{Z} := \{\mathbf{s}_1 + \mathbf{z} : \mathbf{s}_1 \in \mathcal{S}_1, \mathbf{z} \in \mathcal{Z}\},$$

$$\mathcal{S}_1 \ominus \mathcal{Z} := \{\mathbf{s}_1 \in \mathcal{S}_1 : \mathbf{s}_1 + \mathbf{z} \in \mathcal{S}_1, \forall \mathbf{z} \in \mathcal{Z}\},$$

$$\mathbf{P} \circ \mathcal{S}_2 := \{\mathbf{P}\mathbf{s}, \mathbf{s} \in \mathcal{S}_2\}.$$

Definition 1: [24] Consider a dynamical system:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, \mathbf{d}), \quad \mathbf{x} \in \mathcal{X}, \quad \mathbf{u} \in \mathcal{U}, \quad \mathbf{d} \in \mathcal{D},$$

with initial condition $\mathbf{x}(0) \in \mathcal{X}$ and external disturbances $\mathbf{d} \in \mathcal{D}$. A set $\mathcal{X}' \subseteq \mathcal{X}$ is a *Robust Control Invariant (RCI) set* for the system, if there exists a feedback control law $\mathbf{u} := \kappa(\mathbf{x}) \in \mathcal{U}$, such that for all $\mathbf{x}(0) \in \mathcal{X}'$ and for all $\mathbf{d} \in \mathcal{D}$ it holds that $\mathbf{x}(t) \in \mathcal{X}'$ for all $t \geq 0$, along every solution $\mathbf{x}(t)$ of the closed-loop system.

III. PROBLEM STATEMENT

In this section, the overall problem is formulated. Initially, the nominal model of the under-actuated underwater vehicle and the constraints that must be fulfilled are presented.

A. Mathematical Modeling

The prior step before analyzing the proposed methodology is the presentation of the preliminary aspects of the modeling of underwater vehicles. The pose vector of the vehicle with respect to (w.r.t.) the inertial frame $\mathcal{I}$ is denoted by $\boldsymbol{\eta} = [\boldsymbol{\eta}_1^T \boldsymbol{\eta}_2^T]^T \in \mathbb{R}^6$ including the position (i.e., $\boldsymbol{\eta}_1 = [x \ y \ z]^T$) and orientation (i.e., $\boldsymbol{\eta}_2 = [\phi \ \theta \ \psi]^T$) vectors. The $\mathbf{v} = [\mathbf{v}_1^T \ \mathbf{v}_2^T]^T \in \mathbb{R}^6$ is the velocity vector of the vehicle expressed in body fixed frame $\mathcal{V}$ and includes the linear (i.e., $\mathbf{v}_1 = [u \ v \ w]^T$) and angular (i.e., $\mathbf{v}_2 = [p \ q \ r]^T$) velocity vectors where the components have been named according

车辆的车载传感器（例如，多波束成像声呐、车载视觉系统、在线洋流估算器）。在这些情况下，水下车辆必须根据可能的环境变化（即新的障碍物、其他车辆或附近作业的人类等）在线重新计算其路径。有关水下机器人路径规划中开放问题的更多细节，可以参考文献[22]和[23]。

受到上述考虑的驱动，本文提出了一种用于在包括障碍物的受限工作空间中运行的欠驱动自主水下航行器（AUVs）的稳健轨迹跟踪控制方案。具体而言，提出了一种稳健的非线性模型预测控制（NMPC）方案用于欠驱动AUV（即仅在前进、升降和偏航方向上受控的车辆）。在控制设计过程中考虑了各种约束条件，如稀疏障碍物、工作空间边界和控制输入饱和。控制器的目的是引导欠驱动AUV沿着受限且动态变化的工作空间内的期望轨迹行驶。由于工作空间的信息通过车辆的机载传感器不断在线更新，如果更新后的环境变化（即新检测到的障碍物）与参考轨迹存在冲突，机器人将在线重新计算其路径。特别是，通过考虑覆盖系统体积的球体

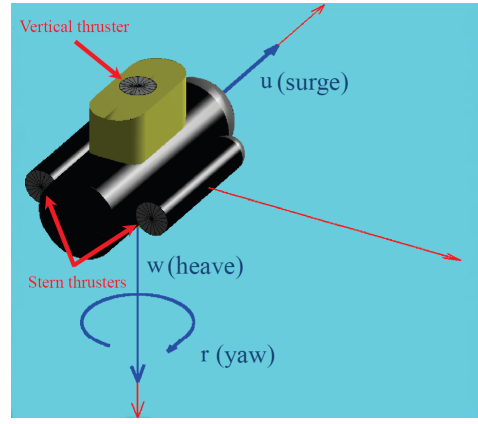


图 1: 欠驱动水下车辆。蓝色表示被驱动的自由度。

在主对角线上有矩阵 $P_1, \dots, P_n$ 的块对角矩阵；

$$\mathcal{B}(c, r) := \{x \in \mathbb{R}^n : \|x - c\|_2 \leq r\},$$

表示在 $\mathbb{R}^n$ 中以 $c \in \mathbb{R}^n$ 为中心、 $r > 0$ 为半径的球，分别为 0。给定集合 $S_1, Z \subseteq \mathbb{R}^n, S_2 \subseteq \mathbb{R}^m$ 和矩阵 $P \in \mathbb{R}^{n \times m}$ ，Minkowski addition、Pontryagin difference 和 matrix-set multiplication 分别定义如下：

$$\begin{aligned} S_1 \oplus Z &:= \{s_1 + z : s_1 \in S_1, z \in Z\}, \\ S_1 \ominus Z &:= \{s_1 \in S_1 : s_1 + z \in S_1, \forall z \in Z\}, \\ P \circ S_2 &:= \{Ps, s \in S_2\}. \end{aligned}$$

定义 1: [24] 考虑一个动力系统:

$$\dot{x} = f(x, u, d), \quad x \in \mathcal{X}, \quad u \in \mathcal{U}, \quad d \in \mathcal{D},$$

初始条件为 $x(0) \in \mathcal{X}$ 并且存在外部扰动 $d \in \mathcal{D}$ 。如果存在一个反馈控制律 $u := \kappa(x) \in \mathcal{U}$ ，使得对所有 $x(0) \in \mathcal{X}'$ 和对所有 $d \in \mathcal{D}$ 都成立 $x(t) \in \mathcal{X}'$ 对所有 $t \geq 0$ ，对于闭环系统的每一条解 $x(t)$ ，则集合 $\mathcal{X}' \subseteq \mathcal{X}$ 是系统的 Robust Control Invariant (RCI) set。

三、问题陈述

在本节中，提出了总体问题的公式化。首先，介绍了欠驱动水下车辆的名义模型以及必须满足的约束条件。

A. Mathematical Modeling

在分析所提出的方法之前的步骤是介绍水下车辆建模的初步方面。车辆相对于惯性坐标系 $\mathcal{I}$ 的位姿向量记为 $\eta = [\eta_1^T \ \eta_2^T]^T \in \mathbb{R}^6$ ，包括位置（即 $\eta_1 = [x \ y \ z]^T$ ）和姿态（即 $\eta_2 = [\phi \ \theta \ \psi]^T$ ）向量。 $v = [v_1^T \ v_2^T]^T \in \mathbb{R}^6$ 是车辆在机体固定坐标系 $\mathcal{V}$ 中表示的速度向量，包括线速度（即 $v_1 = [u \ v \ w]^T$ ）和角速度（即 $v_2 = [p \ q \ r]^T$ ）向量，其中各分量已命名为

II. 符号与背景

在这项工作中，向量用小写粗体字表示，而矩阵用大写粗体字表示。分别用 $\mathbb{N}$ 和 $\mathbb{R}$ 定义正整数集和实数集。给定一个集合 S ，分别用 $|S|$ 和 $S^n := S \times \dots \times S$ 表示它的基数和它的 n 次笛卡尔积。给定一个向量 $z \in \mathbb{R}^n$ ，定义为：

$$\|z\|_2 := \sqrt{z^T z}, \quad \|z\|_\infty := \max_{i=1, \dots, n} |z_i|, \quad \|z\|_P := \sqrt{z^T P z},$$

它分别是欧几里得的、无限的和 P 加权范数，其中 $P \geq 0$ 。符号 $\lambda_{\min}(P)$ 代表 $P \in \mathbb{R}^{n \times n}$ 的特征值实部的最小绝对值； $0_{m \times n} \in \mathbb{R}^{m \times n}$ 和 $I_n \in \mathbb{R}^{n \times n}$ 分别代表所有元素为零的 $m \times n$ 矩阵和单位矩阵。符号 $\text{diag}\{P_1, \dots, P_n\}$ 代表

to SNAME [25] as surge, sway, heave, roll, pitch and yaw respectively (Fig.1). Without loss of generality, according to the standard underwater vehicles' modeling properties [26], assuming that the vehicle is operating at relative low speeds, the dynamic equations of the vehicle can be given as:

$$\dot{\eta} = J(\eta) v \quad (1a)$$

$$M\dot{v} + C(v)v + D(v)v + g(\eta) = \tau \quad (1b)$$

where:

- $\tau = [X, Y, Z, K, M, N]^T \in \mathbb{R}^6$ is the total propulsion force/torque vector (i.e., the body forces and torques generated by the thrusters) applied on the vehicle and expressed in body-fixed frame $\mathcal{V}$;
- $M = M_{RB} + M_A$, where $M_{RB} \in \mathbb{R}^{6 \times 6}$ and $M_A \in \mathbb{R}^{6 \times 6}$ are the inertia matrix for the rigid body and added mass respectively;
- $C(v) = C_{RB}(v) + C_A(v)$, where $C_{RB}(v) \in \mathbb{R}^{6 \times 6}$ and $C_A(v) \in \mathbb{R}^{6 \times 6}$ are the coriolis and centripetal matrix for the rigid body and added mass respectively;
- $D(v) = D_{quad}(v) + D_{lin}(v)$, where $D_{quad}(v) \in \mathbb{R}^{6 \times 6}$ and $D_{lin}(v) \in \mathbb{R}^{6 \times 6}$ are the quadratic and linear drag matrix respectively;
- $g(\eta) \in \mathbb{R}^6$ is the hydrostatic restoring force vector;
- $J(\eta) = \begin{bmatrix} J_1(\eta_2) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & J_2(\eta_2) \end{bmatrix}$ is the Jacobian matrix transforming the velocities from the body-fixed ($\mathcal{V}$) to the inertial ($\mathcal{I}$) frame, in which $J_1(\eta_2) \in SO(3)$ is the well known rotation matrix and $J_2(\eta_2) \in \mathbb{R}^{3 \times 3}$ denotes the lumped transformation matrix [26];

We assume that the vehicle is equipped with three thrusters, which are effective only in surge, heave and yaw motion (Fig.1), meaning that the vehicle is under-actuated along the sway axis. Moreover, we assume that the vehicle is designed with meta-centric restoring forces in order to regulate roll and pitch angles. Thus, the angles ϕ , θ and angular velocities p and q are negligible and we can consider them to be equal to zero. Thus, from now on, we denote by $x = [x, y, z, \psi]^T$ and $v = [u, w, r]^T$ the state and the control input of the system respectively. The vehicle is symmetric around the $x - z$ plane and close to symmetric around the $y - z$ plane. Therefore, we can safely assume that motions in heave, roll and pitch are decoupled [26], [27]. Because the vehicle is operating at relatively low speeds, the coupling effects are considered to be negligible. Due to the above assumptions and the relative low speed of the vehicle, we consider the vehicle's motion equations, which given as follows:

$$\dot{x} = f(x)v + g(x, v) \quad (2)$$

where $\dot{x} = [\dot{x}, \dot{y}, \dot{z}, \dot{\psi}]^T$ and:

$$f(x)v = \begin{bmatrix} \cos(\psi) & 0 & 0 \\ \sin(\psi) & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ w \\ r \end{bmatrix}, \quad g(x, v) = \begin{bmatrix} -\sin(\psi) \\ \cos(\psi) \\ 0 \\ 0 \end{bmatrix} v$$

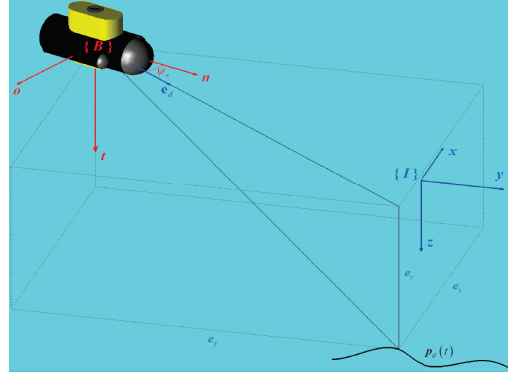


Fig. 2: Graphical illustration of the error and the sea current definitions.

Notice that the v indicates the vehicle velocity in the sway direction. In [28], using Input-to-State Stability (ISS) framework, it was shown that for any vehicle described by (2) and for any bounded control input $[u, r]$ the velocity around the sway direction v can be seen as a bounded perturbation with upper bound $\|v\| \leq \bar{v}$. Consequently, this point is an equilibrium of the kinematic system of (2). Note that throughout this paper the notation $(\bar{\cdot})$ will denote the upper bound for each of the variables. Therefore we consider the system:

$$\dot{x} = f(x)v \quad (3)$$

as the nominal kinematic system of the underwater vehicle, while the function $g(\cdot)$ is considered as a bounded inner disturbance of the system that vanishes at the origin. Also, $g(x, v) \in \Gamma \subset \mathbb{R}^4$ with Γ being a compact set, such that:

$$\|g(x, v)\| \leq \bar{\gamma} \quad \text{with} \quad \bar{\gamma} \triangleq \bar{v} \quad (4)$$

The robot moves under the influence of an irrotational current which behaves as an external disturbance. The current has components with respect to the x , y and z axes, denoted by δ_x , δ_y and δ_z , respectively. Also it is assumed to have a slowly-varying velocity δ_c which is upper bounded by $\|\delta_c\| \leq \bar{\delta}_c$ and it has direction β in the $x - y$ plane and α with respect to the z -axis of the global frame, see Fig. 2. In particular we set $\delta = [\delta_x, \delta_y, \delta_z, 0]^T \in \Delta \subset \mathbb{R}^4$ with Δ being a compact set, where:

$$\begin{aligned} \delta_x &\triangleq \delta_c \cos(\beta) \sin(\alpha) \\ \delta_y &\triangleq \delta_c \sin(\beta) \sin(\alpha) \\ \delta_z &\triangleq \delta_c \cos(\alpha) \end{aligned} \quad (5)$$

It is easy to show that there exists a $\bar{\delta} > 0$ such that $\|\delta\| \leq \bar{\delta}$. Taking into consideration the aforementioned disturbances that affect the vehicle, we can model the perturbed system as follows:

$$\dot{x} = f(x)v + \omega \quad (6)$$

with $\omega = [\omega_1, \omega_2, \omega_3, 0]^T := g(x, v) + \delta \in \Omega \subset \mathbb{R}^4$ as the result of adding the inner and external disturbances of the system and Ω is a compact set with $\Omega = \Delta \oplus \Gamma$. Since the sets Δ and Γ are compact, we have that Ω is also a compact

对于 SNAME [25], 分别表示为冲、横摇、纵荡、横滚、俯仰和偏航 (图 1)。在不失一般性的情况下, 根据标准水下航行器的建模特性 [26], 假设航行器在相对低速下运行, 其动力学方程可表示为:

$$\dot{\eta} = J(\eta) v \quad (1a)$$

$$M\dot{v} + C(v)v + D(v)v + g(\eta) = \tau \quad (1b)$$

在哪里:

- $\tau = [X, Y, Z, K, M, N]^T \in \mathbb{R}^6$ 是作用在飞行器上的总推进力/力矩向量 (即由推进器产生的机体力和力矩), 并以机体固定坐标系 $\mathcal{V}$ 表示;
- $M = M_{RB} + M_A$, 其中 $M_{RB} \in \mathbb{R}^{6 \times 6}$ 和 $M_A \in \mathbb{R}^{6 \times 6}$ 分别是刚体和附加质量的惯性矩阵;
- $C(v) = C_{RB}(v) + C_A(v)$, 其中 $C_{RB}(v) \in \mathbb{R}^{6 \times 6}$ 和 $C_A(v) \in \mathbb{R}^{6 \times 6}$ 分别是刚体和附加质量的科里奥利矩阵和向心矩阵;
- $D(v) = D_{quad}(v) + D_{lin}(v)$, 其中 $D_{quad}(v) \in \mathbb{R}^{6 \times 6}$ 和 $D_{lin}(v) \in \mathbb{R}^{6 \times 6}$ 分别是二次和线性阻力矩阵;
- $g(\eta) \in \mathbb{R}^6$ 是静水恢复力矢量;
- $J(\eta) = \begin{bmatrix} J_1(\eta_2) & 0_{3 \times 3} \\ 0_{3 \times 3} & J_2(\eta_2) \end{bmatrix}$ 是将速度从机体固定系 ($\mathcal{V}$) 转换到惯性系 ($\mathcal{I}$) 的雅可比矩阵, 其中 $J_1(\eta_2) \in SO(3)$ 是众所周知的旋转矩阵, $J_2(\eta_2) \in \mathbb{R}^{3 \times 3}$ 表示汇总的变换矩阵 [26];

我们假设该车辆配备了三个推进器, 这些推进器仅在纵向 (surge)、垂向 (heave) 和偏航 (yaw) 运动中有效 (图1), 这意味着车辆在横向 (sway) 轴方向上是欠驱动的。此外, 我们假设车辆设计了元心回复力以调节横滚 (roll) 和俯仰 (pitch) 角。因此, 角度 ϕ 、 θ 以及角速度 p 和 q 可以忽略不计, 可以认为其等于零。因此, 从现在起, 我们分别用 $x = [x, y, z, \psi]^T$ 和 $v = [u, w, r]^T$ 来表示系统的状态和控制输入。车辆在 $x-z$ 平面上是对称的, 并且在 $y-z$ 平面上接近对称。因此, 我们可以安全地假设垂向、横滚和俯仰的运动是解耦的 [26], [27]。由于车辆以相对较低的速度运行, 耦合效应可以忽略不计。基于上述假设以及车辆的相对低速性, 我们考虑车辆的

$$\dot{x} = f(x)v + g(x, v) \quad (2)$$

在哪里 $\dot{x} = [\dot{x}, \dot{y}, \dot{z}, \dot{\psi}]^T$ 和:

$$f(x)v = \begin{bmatrix} \cos(\psi) & 0 & 0 \\ \sin(\psi) & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ w \\ r \end{bmatrix}, \quad g(x, v) = \begin{bmatrix} -\sin(\psi) \\ \cos(\psi) \\ 0 \\ 0 \end{bmatrix} v$$

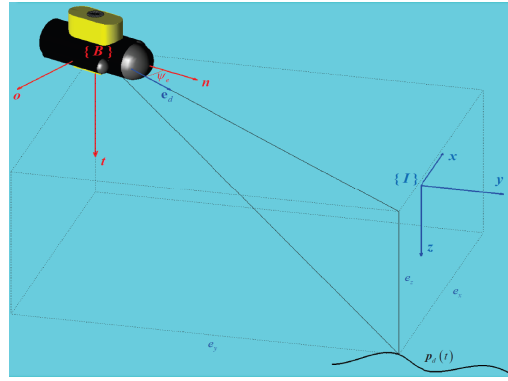


图 2: 错误和海流定义的图示。

请注意, v 表示车辆在横摆方向的速度。在[28]中, 使用输入到状态稳定性 (ISS) 框架显示, 对于由(2)描述的任何车辆以及任何有界控制输入 $[u, r]$, 横摆方向的速度 v 可以视为一个上界为 $\|v\| \leq \bar{v}$ 的有界扰动。因此, 该点是(2)的运动学系统的一个平衡点。请注意, 在本文中, 符号 $(\cdot)$ 将表示每个变量的上界。因此, 我们考虑系统:

$$\dot{x} = f(x)v \quad (3)$$

作为水下车辆的名义运动系统, 而函数 $g(\cdot)$ 被视为系统的有界内部扰动, 在原点消失。同时, $g(x, v) \in \Gamma \subset \mathbb{R}^4$ 与 Γ 为一个紧集, 使得:

$$\|g(x, v)\| \leq \bar{\gamma} \quad \text{with} \quad \bar{\gamma} \triangleq \bar{v} \quad (4)$$

机器人在不可旋转的洋流影响下运动, 该洋流表现为外部扰动。洋流在 x 、 y 和 z 轴方向上有分量, 分别记为 δ_x 、 δ_y 和 δ_z 。同时, 假设洋流具有缓慢变化的速度 δ_c , 其上界为 $\|\delta_c\| \leq \bar{\delta}_c$, 在 $x-y$ 平面内方向为 β , 并相对于全局坐标系的 z 轴的方向为 α , 见图 2。特别地, 我们设 $\delta = [\delta_x, \delta_y, \delta_z, 0]^T \in \Delta \subset \mathbb{R}^4$, 其中 Δ 是一个紧集, 具体如下:

$$\begin{aligned} \delta_x &\triangleq \delta_c \cos(\beta) \sin(\alpha) \\ \delta_y &\triangleq \delta_c \sin(\beta) \sin(\alpha) \\ \delta_z &\triangleq \delta_c \cos(\alpha) \end{aligned} \quad (5)$$

很容易证明存在一个 $\bar{\delta} > 0$, 使得 $\|\delta\| \leq \bar{\delta}$ 。考虑到前面提到的影响车辆的扰动, 我们可以将受扰系统建模如下:

$$\dot{x} = f(x)v + \omega \quad (6)$$

将 $\omega = [\omega_1, \omega_2, \omega_3, 0]^T := g(x, v) + \delta \in \Omega \subset \mathbb{R}^4$ 作为系统内外扰动相加的结果, 并且 Ω 是一个紧集, 且 $\Omega = \Delta \oplus \Gamma$ 。由于集合 Δ 和 Γ 是紧集, 我们有 Ω 也是一个紧集

set, and thus we have:

$$\Omega = \{\boldsymbol{\omega}(t) \in \mathbb{R}^4 : \|\boldsymbol{\omega}(t)\|_2 \leq \bar{\omega}\} \quad (7)$$

with $\bar{\omega} \triangleq \bar{\delta} + \bar{\gamma}$.

B. Geometry of the Workspace

We consider that the underwater vehicle operates inside a workspace $\mathcal{W} \subset \mathbb{R}^3$ with boundary $\partial\mathcal{W}$ and scattered obstacles located within it. Without loss of the generality, the robot and the obstacles are modeled by spheres (i.e., we adopt the spherical world representation [29]). Let $\mathcal{B}(\boldsymbol{\eta}_1, \bar{r})$ be a closed ball that covers the whole vehicle volume (main body and additional equipments). Moreover, let $\mathcal{B}(\boldsymbol{\eta}_1, \bar{R})$ with $\bar{R} > \bar{r}$ be a sensing area where the robot can perceive and update its knowledge of the workspace (i.e., the obstacle locations) using its on-board sensors. Furthermore, the $\mathcal{M}$ static obstacles within the workspace are defined as closed balls described by $\pi_m = \mathcal{B}(\mathbf{p}_{\pi_m}, r_{\pi_m})$, $m \in \{1, \dots, \mathcal{M}\}$, where $\mathbf{p}_{\pi_m} \in \mathbb{R}^3$ is the center and $r_{\pi_m} > 0$ the radius of the obstacle π_m . Additionally, based on the property of spherical world [29], for each pair of obstacles m, m' the following inequality holds:

$$\|\pi_m - \pi_{m'}\| > 2\bar{r} + r_{\pi_m} + r_{\pi_{m'}} \quad (8)$$

which intuitively means that the obstacles m and m' are disjoint in such a way that the entire volume of the vehicle can pass through the free space between them. Therefore, there exists a feasible trajectory $\boldsymbol{\eta}(t)$ for the vehicle such as:

$$\begin{aligned} \mathcal{B}(\boldsymbol{\eta}_1(t), \bar{r}) \cap \{\mathcal{B}(\mathbf{p}_{\pi_m}, r_{\pi_m}) \cup \partial\mathcal{W}\} &= \emptyset, \\ \forall t \geq 0, \forall m \in \{1, \dots, \mathcal{M}\} \end{aligned} \quad (9)$$

C. Constraints

1) *State Constraints*: as already stated, the robot should be able to avoid the newly detected obstacles which may had been unknown to the off-line trajectory planner. This requirement is captured by the state constraint set X of the system, given by:

$$\mathbf{x}(t) \in X \subset \mathbb{R}^4 \quad (10)$$

which in view of (9) can be defined as:

$$\begin{aligned} X := \left\{ \mathbf{x} \in \mathbb{R}^4 : \mathcal{B}(\boldsymbol{\eta}_1, \bar{r}) \cap \{\mathcal{B}(\mathbf{p}_{\pi_m}, r_{\pi_m}) \cup \partial\mathcal{W}\} = \emptyset, \right. \\ \left. \forall m \in \{1, \dots, \mathcal{M}\} \right\} \end{aligned} \quad (11)$$

2) *Input Constraints*: for the needs of several common underwater tasks (e.g., seabed inspection, mosaicking), the vehicle is required to move with relatively low speeds with upper bound denoted by the velocity vector $\bar{\mathbf{v}} = [\bar{u} \ \bar{w} \ \bar{r}]^\top$ which defines the control constraint set V as follows:

$$\mathbf{v}(t) \in V \subseteq \mathbb{R}^3 \quad (12)$$

These constraints are of the form $|u(t)| \leq \bar{u}$, $|w(t)| \leq \bar{w}$, $|r(t)| \leq \bar{r}$, that is:

$$V := \left\{ \mathbf{v} \in \mathbb{R}^3 : \|\mathbf{v}\|_\infty \leq \bar{V} \right\} \quad (13)$$

with $\bar{V} = (\bar{u}^2 + \bar{w}^2 + \bar{r}^2)^{\frac{1}{2}}$.

D. Control Objective

Let $\mathbf{p}_d(t) = [x_d(t), y_d(t), z_d(t)]^T$ denote a smooth desired trajectory with bounded time derivatives. The controller's objective is to guide the underactuated underwater vehicle to track the aforementioned desired trajectory $\mathbf{p}_d(t)$ while guarantying various state and input limitations remain satisfied, despite the presence of exogenous disturbances representing ocean currents and waves. Moreover, as stated previously, the underwater robots in most cases are operating in a partially known and dynamic environment where the knowledge of the workspace (e.g., obstacles positions) is updated as the robot moves through the workspace. Since the desired trajectory for the underwater vehicle is usually derived from an offline path planning technique, it may not be particularly feasible in terms of possible environmental changes. Thus, the controller must have the capability to be flexible with respect to changes on the knowledge of the operating workspace. Hence, the problem of this paper can be defined as follows:

Problem 1: (Robust Tracking Control for an Autonomous Underactuated Underwater Vehicle): Consider an Underactuated Autonomous Underwater Vehicle described by (6) operating in a workspace $\mathcal{W} \subset \mathbb{R}^3$ with state, input constraints as well as disturbances imposed by the sets X , V and Ω as in (10), (12) and (7), respectively. Consider also that the robot and the obstacles are all modeled according to the spherical world representation¹ and the knowledge of the operating workspace $\mathcal{W}$ is constantly updated via the vehicle's on-board sensors inside a sensing region defined by $\mathcal{B}(\boldsymbol{\eta}_1, \bar{R})$. Given a desired trajectory $\mathbf{p}_d(t) = [x_d(t), y_d(t), z_d(t)]^T$, design a feedback control law $\mathbf{v} = \kappa(\mathbf{x})$ such that the desired trajectory $\mathbf{p}_d(t)$ is tracked with guaranteed input and state constraints while avoiding any collision with the obstacles, despite the presence of exogenous disturbances representing ocean currents and waves.

IV. MAIN RESULTS

In this section we present the methodology proposed in order to formulate the solution of Problem-1 defined in Section-III-D. In particular, a Nonlinear Model Predictive Control (NMPC) framework [30]–[33] is utilized, and a relevant robust NMPC analysis, the so-called tube-based approach is provided here for the trajectory tracking problem for underactuated systems in presence of disturbances. The proposed feedback control law consists of two parts: an on-line control law which is the outcome of Finite Horizon Optimal Control Problem (FHOCPP) for the nominal system dynamics and a state feedback law which guarantees that the real system trajectories always lie within a hyper-tube centered along the nominal trajectories. First we begin by defining the error states and the corresponding transformed constraints.

¹as described in section-III-B

set, and thus we have:

$$\Omega = \{\boldsymbol{\omega}(t) \in \mathbb{R}^4 : \|\boldsymbol{\omega}(t)\|_2 \leq \bar{\omega}\} \quad (7)$$

和 $\bar{\omega} \triangleq \bar{\delta} + \bar{\gamma}$ 。

B. Geometry of the Workspace

我们认为水下车辆在一个工作空间 $\mathcal{W} \subset \mathbb{R}^3$ 内运行, 该空间的边界为 $\partial\mathcal{W}$, 并且其中散布有障碍物。在不失一般性的情况下, 机器人和障碍物都被建模为球体 (即, 我们采用球形世界表示 [29])。令 $\mathcal{B}(\boldsymbol{\eta}_1, \bar{r})$ 为覆盖整个车辆体积 (主机体及附加设备) 的闭球。此外, 令 $\mathcal{B}(\boldsymbol{\eta}_1, \bar{R})$ 与 $\bar{R} > \bar{r}$ 为一个感知区域, 机器人可以使用其车载传感器感知并更新对工作空间 (即障碍物位置) 的知识。此外, 工作空间内的 $\mathcal{M}$ 个静态障碍物被定义为由 $\pi_m = \mathcal{B}(\mathbf{p}_{\pi_m}, r_{\pi_m})$, $m \in \{1, \dots, \mathcal{M}\}$ 描述的闭球, 其中 $\mathbf{p}_{\pi_m} \in \mathbb{R}^3$ 是障碍物 π_m 的中心, $r_{\pi_m} > 0$ 是障碍物 π_m 的半径。此外, 基于球形世界的属性 [29], 对于每对障碍物 m, m' , 以下不等式成立:

$$\|\pi_m - \pi_{m'}\| > 2\bar{r} + r_{\pi_m} + r_{\pi_{m'}} \quad (8)$$

直观地, 这意味着障碍物 m 和 m' 是互不相交的, 以至于车辆的整个体积可以通过它们之间的自由空间。因此, 存在一条可行的车辆轨迹 $\boldsymbol{\eta}(t)$, 使得:

$$\begin{aligned} \mathcal{B}(\boldsymbol{\eta}_1(t), \bar{r}) \cap \{\mathcal{B}(\mathbf{p}_{\pi_m}, r_{\pi_m}) \cup \partial\mathcal{W}\} &= \emptyset, \\ \forall t \geq 0, \forall m \in \{1, \dots, \mathcal{M}\} \end{aligned} \quad (9)$$

C. Constraints

1) *State Constraints*: 如前所述, 机器人应能够避开新检测到的障碍物, 这些障碍物可能对离线轨迹规划器是未知的。该要求由系统的状态约束集 X 表达, 具体为:

$$\mathbf{x}(t) \in X \subset \mathbb{R}^4 \quad (10)$$

根据(9)可以定义为:

$$\begin{aligned} X := \left\{ \mathbf{x} \in \mathbb{R}^4 : \mathcal{B}(\boldsymbol{\eta}_1, \bar{r}) \cap \{\mathcal{B}(\mathbf{p}_{\pi_m}, r_{\pi_m}) \cup \partial\mathcal{W}\} = \emptyset, \right. \\ \left. \forall m \in \{1, \dots, \mathcal{M}\} \right\} \end{aligned} \quad (11)$$

2) *Input Constraints*: 对于若干常见的水下任务 (例如, 海床检查、图像拼接) 的需求, 车辆需要以相对较低的速度移动, 其上限由速度向量 $\bar{\mathbf{v}} = [\bar{u} \ \bar{w} \ \bar{r}]^\top$ 表示, 该速度向量定义了控制约束集 V , 如下所示:

$$\mathbf{v}(t) \in V \subseteq \mathbb{R}^3 \quad (12)$$

这些约束的形式为 $|u(t)| \leq \bar{u}$ 、 $|w(t)| \leq \bar{w}$ 、 $|r(t)| \leq \bar{r}$, 也就是说:

$$V := \left\{ \mathbf{v} \in \mathbb{R}^3 : \|\mathbf{v}\|_\infty \leq \bar{V} \right\} \quad (13)$$

使用 $\bar{V} = (\bar{u}^2 + \bar{w}^2 + \bar{r}^2)^{\frac{1}{2}}$ 。

D. Control Objective

令 $\mathbf{p}_d(t) = [x_d(t), y_d(t), z_d(t)]^T$ 表示具有有界时间导数的光滑期望轨迹。控制器的目标是引导欠驱动水下车辆跟踪上述期望轨迹 $\mathbf{p}_d(t)$, 同时保证在存在表示洋流和波浪的外部干扰的情况下, 各种状态和输入限制仍然得到满足。此外, 如前所述, 大多数情况下水下机器人在部分已知且动态的环境中运行, 其中工作空间的知识 (例如障碍物位置) 会随着机器人在工作空间中的移动而更新。由于水下车辆的期望轨迹通常是由离线路径规划技术得出的, 因此在可能的环境变化方面, 它可能并不是特别可行。因此, 控制器必须具备对操作工作空间知识变化的灵活适应能力。因此, 本文的问题可以定义如下

Problem 1: (自主欠驱动水下车辆的鲁棒跟踪控制: 考虑一个由(6)描述的欠驱动自主水下车辆, 在工作空间 $\mathcal{W} \subset \mathbb{R}^3$ 中运行, 其状态、输入约束以及由集合 X 、 V 和 Ω 分别在(10)、(12)和(7)中给出的扰动施加。还考虑机器人和障碍物均按照球形世界表示¹建模, 并且通过车辆的车载传感器在由 $\mathcal{B}(\boldsymbol{\eta}_1, \bar{R})$ 定义的感知区域内不断更新对工作空间 $\mathcal{W}$ 的认知。给定期望轨迹 $\mathbf{p}_d(t) = [x_d(t), y_d(t), z_d(t)]^T$, 设计反馈控制律 $\mathbf{v} = \kappa(\mathbf{x})$, 使得在保证输入和状态约束的同时, 能够跟踪期望轨迹 $\mathbf{p}_d(t)$, 并且在存在代表洋流和海浪的外部扰动的情况下, 避免与障碍物发生碰撞。

四、主要结果

在本节中, 我们提出了解决第三节D部分中定义的问题1的方法论。具体而言, 采用了非线性模型预测控制 (NMPC) 框架 [30]–[33], 并提供了相关的稳健NMPC分析, 即所谓的基于管道的方法, 用于在存在扰动的情况下对欠驱动系统的轨迹跟踪问题。所提出的反馈控制律由两部分组成: 一部分是在线控制律, 它是名义系统动力学的有限视域最优控制问题 (FHOC) 的结果; 另一部分是状态反馈律, 它保证实际系统轨迹始终位于沿名义轨迹中心的超管道内。首先, 我们从定义误差状态及相应的变换约束开始。

如第III-B节所述

A. Error Definitions

Given the desired trajectory $\mathbf{p}_d(t) = [x_d(t), y_d(t), z_d(t)]^T$, let us define the position errors:

$$e_x(t) = x - x_d(t), \quad e_y(t) = y - y_d(t), \quad e_z(t) = z - z_d(t) \quad (14)$$

the projected on the horizontal plane distance error:

$$e_d(t) = \sqrt{e_x^2(t) + e_y^2(t)} \quad (15)$$

as well as the projected on the horizontal plane orientation error:

$$e_o(t) = \frac{e_x(t)}{e_d(t)} s_{\psi(t)} - \frac{e_y(t)}{e_d(t)} c_{\psi(t)} \quad (16)$$

where $s_\star = \sin(\star)$ and $c_\star = \cos(\star)$. It should be noted that the tracking control problem is solved if the projected on the horizontal plane distance error e_d , the vertical error e_z and the orientation error e_o reduce to zero. Moreover it should be noticed that the orientation error e_o is well-defined only for nonzero values of e_d . For this respect, the proposed approach has been designed to further guarantee that $e_d(t) \geq \epsilon, \forall t \geq 0$, where ϵ is an arbitrarily small positive value, that avoids the aforementioned singularity issue when $e_d \rightarrow 0$. Therefore, in view of (15) and considering the state constraint set X of (10), we can define a feasible error set given as:

$$\mathcal{E} := \{\mathbf{x} \in X : \sqrt{e_x^2(t) + e_y^2(t)} \geq \epsilon, \text{ with } \epsilon > 0\} \quad (17)$$

Now, differentiating the aforementioned errors of (14)-(16) and employing (6), we arrive at:

$$\dot{e}_d = \frac{e_x c_{\psi} + e_y s_{\psi}}{e_d} u - \frac{e_x \dot{x}_d + e_y \dot{y}_d}{e_d} + \frac{e_x \omega_1 + e_y \omega_2}{e_d} \quad (18)$$

$$\dot{e}_z = w - \dot{z}_d + \omega_3 \quad (19)$$

$$\begin{aligned} \dot{e}_o = & \frac{e_y s_{\psi} c_{\psi}^2 - e_x s_{\psi}^2 c_{\psi}}{e_d^2} u + \frac{e_x e_y}{e_d^2} r + \frac{e_y \omega_1 - e_x \omega_2}{e_d^2} s_{\psi} c_{\psi} \\ & + \frac{(e_x \dot{y}_d - e_y \dot{x}_d) s_{\psi} c_{\psi}}{e_d^2} \end{aligned} \quad (20)$$

By defining the error vector $\mathbf{e} = [e_d, e_z, e_o]^T$, the aforementioned formulas can be written in matrix form as:

$$\dot{\mathbf{e}} = J(\mathbf{e}, \mathbf{p}_d) \mathbf{v} + \zeta(\mathbf{e}, \dot{\mathbf{p}}_d) + \xi(\mathbf{e}, \boldsymbol{\omega}) \quad (21)$$

where:

$$\begin{aligned} J(\mathbf{e}, \mathbf{p}_d) \mathbf{v} := & \begin{bmatrix} \frac{e_x c_{\psi} + e_y s_{\psi}}{e_d} & 0 & 0 \\ 0 & 1 & 0 \\ \frac{e_y s_{\psi} c_{\psi}^2 - e_x s_{\psi}^2 c_{\psi}}{e_d^2} & 0 & \frac{e_x e_y}{e_d^2} \end{bmatrix} \begin{bmatrix} u \\ w \\ r \end{bmatrix} \\ \zeta(\mathbf{e}, \dot{\mathbf{p}}_d) := & \begin{bmatrix} -\frac{e_x \dot{x}_d + e_y \dot{y}_d}{e_d} \\ -\dot{z}_d \\ \frac{(e_x \dot{y}_d - e_y \dot{x}_d) s_{\psi} c_{\psi}}{e_d^2} \end{bmatrix}, \quad \xi(\mathbf{e}, \boldsymbol{\omega}) := \begin{bmatrix} \frac{e_x \omega_1 + e_y \omega_2}{e_d} \\ \omega_3 \\ \frac{e_y \omega_1 - e_x \omega_2}{e_d^2} s_{\psi} c_{\psi} \end{bmatrix} \end{aligned}$$

which are the uncertain error dynamics of the underwater vehicle system. The corresponding nominal error dynamics can be now given by:

$$\dot{\hat{\mathbf{e}}} = J(\hat{\mathbf{e}}, \mathbf{p}_d) \hat{\mathbf{v}} + \zeta(\hat{\mathbf{e}}, \dot{\mathbf{p}}_d) \quad (22)$$

It should be noticed that we use the $\hat{\mathbf{e}}$ notation for the nominal error state in order to account for the mismatch between the real error state and the nominal one which will be used in the following analysis.

Remark 1: It should be noted that the constraint set $\mathcal{E}$ in (17) guarantees that $J(\cdot)$ is non-singular. Thus, there exists a strictly positive constant $\underline{J}$ such that: $\lambda_{\min} \left(\frac{J(\cdot) + J^T(\cdot)}{2} \right) \geq \underline{J} > 0$.

Remark 2: It can easily be shown that the function $\xi(\mathbf{e}, \boldsymbol{\omega})$ appearing in (21) is a bounded function and there exists a strictly positive constant $\tilde{\xi}$ such that: $\|\xi(\mathbf{e}, \boldsymbol{\omega})\| \leq \tilde{\xi}$, $\tilde{\xi} > 0$. In particular, by noting that $\frac{|e_x|}{\sqrt{e_x^2 + e_y^2}} \leq 1$, $\frac{|e_y|}{\sqrt{e_x^2 + e_y^2}} \leq 1$ and $|s_{\psi}| \leq 1$, $|c_{\psi}| \leq 1$, we deduce that ξ does not depend of the bounds of $\mathbf{e}$.

B. State Feedback Design

Consider the feedback law:

$$\mathbf{v} = \hat{\mathbf{v}}(\hat{\mathbf{e}}) + \kappa(\mathbf{e}, \hat{\mathbf{e}}) \quad (23)$$

which consists of a nominal control action $\hat{\mathbf{v}}(\hat{\mathbf{e}}) \in V$ and a state feedback law $\kappa : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow V$. The control action $\hat{\mathbf{v}}(\hat{\mathbf{e}})$ will be the outcome of a FHOC solved for the nominal dynamics (22) while the state feedback law $\kappa(\cdot)$ is designed in order to guarantee that the real trajectory $\mathbf{e}(t)$ (i.e., the solution of (21)) always remain inside a bounded tube centered along the nominal trajectory $\hat{\mathbf{e}}(t)$ i.e., the solution of (22). Now let us define by $\mathbf{z}(t) = [z_1(t), z_2(t), z_3(t)]^T$, and then the deviation between the real error state and the nominal one is given as:

$$\boldsymbol{\rho}(t) := \mathbf{e}(t) - \hat{\mathbf{e}}(t) \quad (24)$$

with $\boldsymbol{\rho}(0) = \mathbf{e}(0) - \hat{\mathbf{e}}(0) = \mathbf{0}$. In view of (24), the dynamics of $\boldsymbol{\rho}(t)$ can be given as:

$$\begin{aligned} \dot{\boldsymbol{\rho}} &= \dot{\mathbf{e}} - \dot{\hat{\mathbf{e}}} \\ &= J(\mathbf{e}, \mathbf{p}_d) \mathbf{v} - J(\hat{\mathbf{e}}, \mathbf{p}_d) \hat{\mathbf{v}} + \zeta(\mathbf{e}, \dot{\mathbf{p}}_d) - \zeta(\hat{\mathbf{e}}, \dot{\mathbf{p}}_d) + \xi(\mathbf{e}, \boldsymbol{\omega}). \end{aligned}$$

By adding and subtracting the term $J(\mathbf{e}, \mathbf{p}_d) \hat{\mathbf{v}}$ and by defining the function $h(\mathbf{e}, \hat{\mathbf{v}}) := J(\mathbf{e}, \mathbf{p}_d) \hat{\mathbf{v}}$, the latter becomes:

$$\begin{aligned} \dot{\boldsymbol{\rho}} &= h(\mathbf{e}, \hat{\mathbf{v}}) - h(\hat{\mathbf{e}}, \hat{\mathbf{v}}) + J(\mathbf{e}, \mathbf{p}_d) (\mathbf{v} - \hat{\mathbf{v}}) \\ &\quad + \zeta(\mathbf{e}, \dot{\mathbf{p}}_d) - \zeta(\hat{\mathbf{e}}, \dot{\mathbf{p}}_d) + \xi(\mathbf{e}, \boldsymbol{\omega}). \end{aligned} \quad (25)$$

Note that for the continuously differentiable functions $h(\cdot)$ and $\zeta(\cdot)$ the following hold:

$$\|h(\mathbf{e}, \hat{\mathbf{v}}) - h(\hat{\mathbf{e}}, \hat{\mathbf{v}})\| \leq \mathcal{L}_1 \|\mathbf{e} - \hat{\mathbf{e}}\| = \mathcal{L}_1 \|\boldsymbol{\rho}\|, \quad (26a)$$

$$\|\zeta(\mathbf{e}, \dot{\mathbf{p}}_d) - \zeta(\hat{\mathbf{e}}, \dot{\mathbf{p}}_d)\| \leq \mathcal{L}_2 \|\mathbf{e} - \hat{\mathbf{e}}\| = \mathcal{L}_2 \|\boldsymbol{\rho}\|, \quad (26b)$$

where $\mathcal{L}_1, \mathcal{L}_2 > 0$ stand for their Lipschitz constants. Now based on the aforementioned analysis the following Lemma can be stated:

Lemma 1: The state feedback law designed by:

$$\kappa(\mathbf{e}, \hat{\mathbf{e}}) := -\sigma(\mathbf{e} - \hat{\mathbf{e}}), \quad (27)$$

where σ is chosen such that:

$$\sigma > \frac{\mathcal{L}_1 + \mathcal{L}_2}{\underline{J}}. \quad (28)$$

A. Error Definitions

给定期望轨迹 $\mathbf{p}_d(t) = [x_d(t), y_d(t), z_d(t)]^T$ ，我们定义位置误差为：

$$e_x(t) = x - x_d(t), e_y(t) = y - y_d(t), e_z(t) = z - z_d(t) \quad (14)$$

在水平面上的投影距离误差：

$$e_d(t) = \sqrt{e_x^2(t) + e_y^2(t)} \quad (15)$$

以及水平面投影的方向误差预测：

$$e_o(t) = \frac{e_x(t)}{e_d(t)} s_\psi(t) - \frac{e_y(t)}{e_d(t)} c_\psi(t) \quad (16)$$

其中 $s_\star = \sin(\star)$ 和 $c_\star = \cos(\star)$ 。需要注意的是，如果在水平面上的投影距离误差 e_d 、垂直误差 e_z 和方向误差 e_o 减少到零，则跟踪控制问题就得以解决。此外需要注意的是，方向误差 e_o 仅在 e_d 非零值时定义良好。为此，所提出的方法已经被设计以进一步保证 $e_d(t) \geq \epsilon, \forall t \geq 0$ ，其中 ϵ 是任意小的正数，从而在 $e_d \rightarrow 0$ 时避免上述奇异性问题。因此，考虑到 (15) 并结合状态约束集合 X （见 (10)），我们可以定义一个可行误差集合，如下所示：

$$\mathcal{E} := \{\mathbf{x} \in X : \sqrt{e_x^2(t) + e_y^2(t)} \geq \epsilon, \text{ with } \epsilon > 0\} \quad (17)$$

现在，对上述公式 (14)-(16) 的误差进行微分，并使用 (6)，我们得到：

$$\dot{e}_d = \frac{e_x c_\psi + e_y s_\psi}{e_d} u - \frac{e_x \dot{x}_d + e_y \dot{y}_d}{e_d} + \frac{e_x \omega_1 + e_y \omega_2}{e_d} \quad (18)$$

$$\dot{e}_z = w - \dot{z}_d + \omega_3 \quad (19)$$

$$\begin{aligned} \dot{e}_o = & \frac{e_y s_\psi c_\psi^2 - e_x s_\psi^2 c_\psi}{e_d^2} u + \frac{e_x e_y}{e_d^2} r + \frac{e_y \omega_1 - e_x \omega_2}{e_d^2} s_\psi c_\psi \\ & + \frac{(e_x \dot{y}_d - e_y \dot{x}_d) s_\psi c_\psi}{e_d^2} \end{aligned} \quad (20)$$

通过定义误差向量 $\mathbf{e} = [e_d, e_z, e_o]^T$ ，上述公式可以写成矩阵形式如下：

$$\dot{\mathbf{e}} = J(\mathbf{e}, \mathbf{p}_d) \mathbf{v} + \zeta(\mathbf{e}, \dot{\mathbf{p}}_d) + \xi(\mathbf{e}, \boldsymbol{\omega}) \quad (21)$$

在哪里：

$$J(\mathbf{e}, \mathbf{p}_d) \mathbf{v} := \begin{bmatrix} \frac{e_x c_\psi + e_y s_\psi}{e_d} & 0 & 0 \\ 0 & 1 & 0 \\ \frac{e_y s_\psi c_\psi^2 - e_x s_\psi^2 c_\psi}{e_d^2} & 0 & \frac{e_x e_y}{e_d^2} \end{bmatrix} \begin{bmatrix} u \\ w \\ r \end{bmatrix}$$

$$\zeta(\mathbf{e}, \dot{\mathbf{p}}_d) := \begin{bmatrix} -\frac{e_x \dot{x}_d + e_y \dot{y}_d}{e_d} \\ -\dot{z}_d \\ \frac{(e_x \dot{y}_d - e_y \dot{x}_d) s_\psi c_\psi}{e_d^2} \end{bmatrix}, \quad \xi(\mathbf{e}, \boldsymbol{\omega}) := \begin{bmatrix} \frac{e_x \omega_1 + e_y \omega_2}{e_d} \\ \omega_3 \\ \frac{e_y \omega_1 - e_x \omega_2}{e_d^2} s_\psi c_\psi \end{bmatrix}$$

这是水下车辆系统的不确定误差动力学。相应的名义误差动力学现在可以表示为：

$$\dot{\hat{\mathbf{e}}} = J(\hat{\mathbf{e}}, \mathbf{p}_d) \hat{\mathbf{v}} + \zeta(\hat{\mathbf{e}}, \dot{\mathbf{p}}_d) \quad (22)$$

应当注意，我们使用 $\hat{\mathbf{e}}$ 表示名义误差状态，以考虑实际误差状态与将在后续分析中使用的名义状态之间的不匹配。

备注1：应当注意，(17)中的约束集合 $\mathcal{E}$ 保证了 $J(\cdot)$ 是非奇异的。因此，存在一个严格正的常数 $\underline{J}$ ，使得：

$$\lambda_{\min} \left(\frac{J(\cdot) + J^T(\cdot)}{2} \right) \geq \underline{J} > 0。$$

备注2：可以很容易地证明，出现在(21)中的函数 $\xi(\mathbf{e}, \boldsymbol{\omega})$ 是有界函数，并且存在一个严格正的常数 $\tilde{\xi}$ ，使得： $\|\xi(\mathbf{e}, \boldsymbol{\omega})\| \leq \tilde{\xi}, \tilde{\xi} > 0$ 。特别地，通过注意到 $\frac{|e_x|}{\sqrt{e_x^2 + e_y^2}} \leq 1$ ， $\frac{|e_y|}{\sqrt{e_x^2 + e_y^2}} \leq 1$ 和 $|s_\psi| \leq 1, |c_\psi| \leq 1$ ，我们可以推断出 ξ 不依赖于 $\mathbf{e}$ 的界。

B. State Feedback Design

考虑反馈律：

$$\mathbf{v} = \hat{\mathbf{v}}(\hat{\mathbf{e}}) + \kappa(\mathbf{e}, \hat{\mathbf{e}}) \quad (23)$$

它由一个名义控制作用 $\hat{\mathbf{v}}(\hat{\mathbf{e}}) \in V$ 和一个状态反馈律 κ 组成： $\mathbb{R}^3 \times \mathbb{R}^3 \rightarrow V$ 。控制作用 $\hat{\mathbf{v}}(\hat{\mathbf{e}})$ 将是针对名义动力学 (22) 求解的 FHOC 的结果，而状态反馈律 $\kappa(\cdot)$ 则旨在保证实际轨迹 $\mathbf{e}(t)$ （即 (21)）的解，始终保持在以名义轨迹 $\hat{\mathbf{e}}(t)$ ，即 (22) 的解为中心的有界管道内。现在我们定义 $\mathbf{z}(t) = [z_1(t), z_2(t), z_3(t)]^T$ ，然后实际误差状态与名义状态之间的偏差表示为：

$$\boldsymbol{\rho}(t) := \mathbf{e}(t) - \hat{\mathbf{e}}(t) \quad (24)$$

使用 $\boldsymbol{\rho}(0) = \mathbf{e}(0) - \hat{\mathbf{e}}(0) = \mathbf{0}$ 。根据 (24)， $\boldsymbol{\rho}(t)$ 的动力学可以表示为：

$$\begin{aligned} \dot{\boldsymbol{\rho}} = & \dot{\mathbf{e}} - \dot{\hat{\mathbf{e}}} \\ = & J(\mathbf{e}, \mathbf{p}_d) \mathbf{v} - J(\hat{\mathbf{e}}, \mathbf{p}_d) \hat{\mathbf{v}} + \zeta(\mathbf{e}, \dot{\mathbf{p}}_d) - \zeta(\hat{\mathbf{e}}, \dot{\mathbf{p}}_d) + \xi(\mathbf{e}, \boldsymbol{\omega}). \end{aligned}$$

通过加上和减去项 $J(\mathbf{e}, \mathbf{p}_d) \hat{\mathbf{v}}$ 并定义函数 $h(\mathbf{e}, \hat{\mathbf{v}})$ ：

$= J(\mathbf{e}, \mathbf{p}_d) \hat{\mathbf{v}}$ ，后者变为：

$$\begin{aligned} \dot{\boldsymbol{\rho}} = & h(\mathbf{e}, \hat{\mathbf{v}}) - h(\hat{\mathbf{e}}, \hat{\mathbf{v}}) + J(\mathbf{e}, \mathbf{p}_d)(\mathbf{v} - \hat{\mathbf{v}}) \\ & + \zeta(\mathbf{e}, \dot{\mathbf{p}}_d) - \zeta(\hat{\mathbf{e}}, \dot{\mathbf{p}}_d) + \xi(\mathbf{e}, \boldsymbol{\omega}). \end{aligned} \quad (25)$$

请注意，对于连续可微函数 $h(\cdot)$ 和 $\zeta(\cdot)$ ，以下成立：

$$\|h(\mathbf{e}, \hat{\mathbf{v}}) - h(\hat{\mathbf{e}}, \hat{\mathbf{v}})\| \leq \mathcal{L}_1 \|\mathbf{e} - \hat{\mathbf{e}}\| = \mathcal{L}_1 \|\boldsymbol{\rho}\|, \quad (26a)$$

$$\|\zeta(\mathbf{e}, \dot{\mathbf{p}}_d) - \zeta(\hat{\mathbf{e}}, \dot{\mathbf{p}}_d)\| \leq \mathcal{L}_2 \|\mathbf{e} - \hat{\mathbf{e}}\| = \mathcal{L}_2 \|\boldsymbol{\rho}\|, \quad (26b)$$

其中 $\mathcal{L}_1, \mathcal{L}_2 > 0$ 代表它们的 Lipschitz 常数。现在基于上述分析，可以提出以下引理：

引理 1：由下式设计的状态反馈律：

$$\kappa(\mathbf{e}, \hat{\mathbf{e}}) := -\sigma(\mathbf{e} - \hat{\mathbf{e}}), \quad (27)$$

其中 σ 的选择如下：

$$\sigma > \frac{\mathcal{L}_1 + \mathcal{L}_2}{\underline{J}}. \quad (28)$$

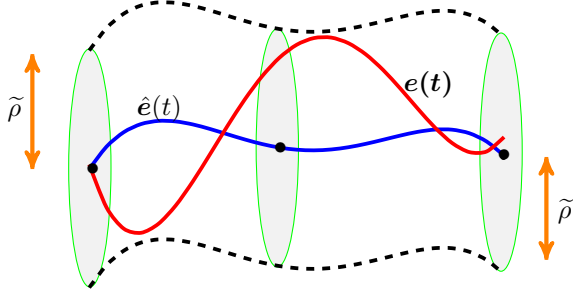


Fig. 3: The hyper-tube centered along the trajectory $\hat{e}(t)$ (depicted by the blue line) with radius $\tilde{\rho}$. Under the proposed control law, the real trajectory $e(t)$ (depicted with red line) lies inside the hyper-tube for all times, i.e., $\|\rho(t)\| \leq \tilde{\rho}, \forall t \in \mathbb{R}_{\geq 0}$.

renders the set:

$$\mathcal{P} := \{\rho \in \mathbb{R}^n : \|\rho\| \leq \tilde{\rho}\},$$

an RCI set for the error dynamics (25), according to Definition 1, with:

$$\tilde{\rho} := \frac{\tilde{\xi}}{\sigma \underline{J} - \mathcal{L}_1 - \mathcal{L}_2} > 0.$$

Proof : Consider the positive definite function $\Lambda(\rho) = \frac{1}{2}\|\rho\|^2$. The time derivative of Λ along the trajectories of the system (25) is given by:

$$\begin{aligned} \dot{\Lambda}(\rho) &= \rho^\top \dot{\rho} \\ &= \rho^\top [h(e, \hat{v}) - h(\hat{e}, \hat{v})] + \rho^\top [\zeta(e, \dot{p}_d) - \zeta(\hat{e}, \dot{p}_d)] \\ &\quad + \rho^\top J(e, p_d)(v - \hat{v}) + \rho^\top \xi(e, \omega). \end{aligned}$$

By employing (26a)-(26b) and the boundedness of $\|\xi(\cdot)\|$ given in Remark 2, i.e., $\|\xi(e, \omega)\| \leq \tilde{\xi}$, $\tilde{\xi} > 0$, the latter becomes:

$$\dot{\Lambda}(\rho) \leq (\mathcal{L}_1 + \mathcal{L}_2)\|\rho\|^2 + \rho^\top J(e, p_d)(v - \hat{v}) + \tilde{\xi}\|\rho\|. \quad (29)$$

Now, by taking $J = \frac{J+J^\top}{2} + \frac{J-J^\top}{2}$; using the fact that $y^\top \frac{J-J^\top}{2} y = 0, \forall y$; and substituting the control law (27), the inequality of (29) yields:

$$\dot{\Lambda}(\rho) \leq [-(\sigma \underline{J} - \mathcal{L}_1 - \mathcal{L}_2)\|\rho\| + \tilde{\xi}]\|\rho\|.$$

Thus, it holds that:

$$\dot{\Lambda}(\rho) < 0 \text{ when } \|\rho\| > \frac{\tilde{\xi}}{\sigma \underline{J} - \mathcal{L}_1 - \mathcal{L}_2}. \quad (30)$$

Moreover, owing to the fact that $\rho(0) = 0$, we have:

$$\|\rho(t)\| \leq \frac{\tilde{\xi}}{\sigma \underline{J} - \mathcal{L}_1 - \mathcal{L}_2}, \forall t \geq 0.$$

which concludes the proof. $\square$

A graphical illustration of the proposed tube based control strategy is given in Fig 3. Under the proposed control scheme (23), the real trajectory $e(t)$ lies inside the hyper-tube which is centered along the nominal trajectory $\hat{e}$ with radius $\tilde{\rho}$ for all times, i.e., $\|\rho(t)\| \leq \tilde{\rho}, \forall t \in \mathbb{R}_{\geq 0}$.

C. Online Optimal Control

As mentioned before, the control action $\hat{v}(\hat{e})$ in eq. (23) will be the outcome of a FHOC solved for the nominal dynamics eq. (22). In this respect, consider a sequence of sampling times $\{t_k\}$, $k \in \mathbb{N}$, with a constant sampling period $0 < \delta_t < T$, where T is a prediction horizon such that $t_{k+1} := t_k + \delta_t, \forall k \in \mathbb{N}$. At each sampling time t_k , a FHOC is solved as follows:

$$\min_{\hat{v}(\cdot)} \left\{ \|\hat{e}(t_k + T)\|_P^2 + \int_{t_k}^{t_k+T} [\|\hat{e}(\mathfrak{s})\|_Q^2 + \|\hat{u}(\mathfrak{s})\|_R^2] d\mathfrak{s} \right\} \quad (31a)$$

subject to:

$$\dot{\hat{e}}(\mathfrak{s}) = J(\hat{e}(\mathfrak{s}), p_d)\hat{v} + \zeta(\hat{e}(\mathfrak{s}), \dot{p}_d), \quad \hat{e}(t_k) = e(t_k), \quad (31b)$$

$$\hat{e}(\mathfrak{s}) \in \bar{\mathcal{E}}, \quad \hat{v}(\mathfrak{s}) \in \bar{V}, \quad \forall \delta_t \in [t_k, t_k + T], \quad (31c)$$

$$\hat{e}(t_k + T) \in \mathcal{F}, \quad (31d)$$

where $Q, P \in \mathbb{R}^{3 \times 3}$ and $R \in \mathbb{R}^{3 \times 3}$ are positive definite gain matrices. Moreover, $\bar{\mathcal{E}}, \bar{V}$ and $\mathcal{F}$ are designing sets that are defined in order to guarantee that while the solution of FHOC (31a)-(31d) is derived for the nominal dynamics (22), the real trajectory $e(t)$ and control inputs $v(t)$ satisfy the corresponding state and input constraint sets $\mathcal{E}$ and V respectively. More specifically, the following modification is performed:

$$\bar{\mathcal{E}} := \mathcal{E} \ominus \mathcal{P}, \quad \bar{V} := V \ominus [-\sigma \circ \mathcal{P}]. \quad (32)$$

This intuitively means that the sets $\mathcal{E}, V$ are tightened accordingly, in order to guarantee that while the nominal states $\hat{e}$ and the nominal control input $\hat{v}$ are calculated, the corresponding real error states e and real control input v satisfy the state and input constraints $\mathcal{E}, \mathcal{P}$ and $\mathcal{U}$, respectively². Define the *terminal set* by:

$$\mathcal{F} := \{\hat{e} \in \bar{\mathcal{E}} : \|\hat{e}\|_P \leq \bar{\epsilon}\}, \quad \bar{\epsilon} > 0, \quad (33)$$

which is employed here in order to enforce the stability of the system [31].

Newly Detected Obstacles: as mentioned before, the obstacles within the workspace may be detected online by the vehicle's on-board sensors (e.g., multi-beam imaging or side scan sonar). In such a case, it should be assured that the solution of the FHOC corresponds to the region that is accessible by the sensing capabilities of the vehicle. This intuitively means it is required any potential new obstacles to be visible by the vehicle even in the worst case (i.e., maximum velocity of the robot under maximum disturbances). Thus, assuming that $\bar{R}$ denotes the sensing range of the system as it is already stated in Section-III-B, the prediction horizon T should be set as follows:

$$T \leq \frac{\bar{R}}{\bar{V} + \tilde{\xi}} \quad (34)$$

where $\bar{V}$ is defined in (13).

Remark 3: It should be noticed that in a real scenario, AUVs use sonar sensors to obtain knowledge about the

²This constitutes a standard constraints set modification technique adopted in tube-based NMPC frameworks. For more details see [34].

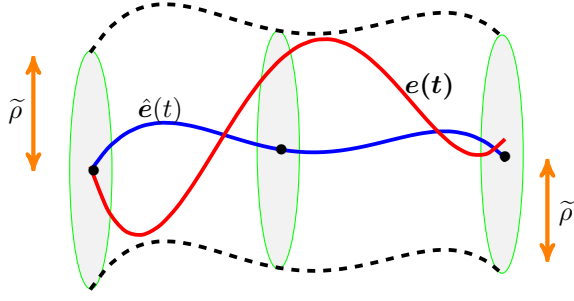


图3: 以轨迹 $\hat{e}(t)$ (为中心的超管道, 由蓝色线条) 表示, 半径为 $\tilde{\rho}$ 。在所提出的控制律下, 实际轨迹 $e(t)$ (由红色线条) 表示, 在所有时间内都位于超管道内部, 即 $\|\rho(t)\| \leq \tilde{\rho}, \forall t \in \mathbb{R}_{\geq 0}$ 。

呈现集合:

$$\mathcal{P} := \{\rho \in \mathbb{R}^n : \|\rho\| \leq \tilde{\rho}\},$$

根据定义1, 针对误差动力学(25)的一个RCI集合, 如下所示:

$$\tilde{\rho} := \frac{\tilde{\xi}}{\sigma \underline{J} - \mathcal{L}_1 - \mathcal{L}_2} > 0.$$

证明: 考虑正定函数 $\Lambda(\rho) = \frac{1}{2}\|\rho\|^2$ 。沿系统 (25) 的轨迹, Λ 的时间导数为:

$$\begin{aligned} \dot{\Lambda}(\rho) &= \rho^\top \dot{\rho} \\ &= \rho^\top [h(e, \hat{v}) - h(\hat{e}, \hat{v})] + \rho^\top [\zeta(e, \dot{p}_d) - \zeta(\hat{e}, \dot{p}_d)] \\ &\quad + \rho^\top J(e, p_d)(v - \hat{v}) + \rho^\top \xi(e, \omega). \end{aligned}$$

通过使用 (26a)-(26b) 以及在备注 2 中给出的 $\|\xi(\cdot)\|$ 的有界性, 即 $\|\xi(e, \omega)\| \leq \tilde{\xi}, \tilde{\xi} > 0$, 后者变为:

$$\dot{\Lambda}(\rho) \leq (\mathcal{L}_1 + \mathcal{L}_2)\|\rho\|^2 + \rho^\top J(e, p_d)(v - \hat{v}) + \tilde{\xi}\|\rho\|. \quad (29)$$

现在, 通过取 $J = \frac{J+J^\top}{2} + \frac{J-J^\top}{2}$; 利用 $y^\top \frac{J-J^\top}{2} y = 0, \forall y$; 并代入控制律 (27), 不等式 (29) 得到:

$$\dot{\Lambda}(\rho) \leq [-(\sigma \underline{J} - \mathcal{L}_1 - \mathcal{L}_2)\|\rho\| + \tilde{\xi}]\|\rho\|.$$

因此, 它成立如下:

$$\dot{\Lambda}(\rho) < 0 \text{ when } \|\rho\| > \frac{\tilde{\xi}}{\sigma \underline{J} - \mathcal{L}_1 - \mathcal{L}_2}. \quad (30)$$

此外, 由于 $\rho(0) = 0$, 我们有:

$$\|\rho(t)\| \leq \frac{\tilde{\xi}}{\sigma \underline{J} - \mathcal{L}_1 - \mathcal{L}_2}, \forall t \geq 0.$$

这就结束了证明。□

图3给出了所提出的基于管道的控制策略的图示。在所提出的控制方案(23)下, 实际轨迹 $e(t)$ 位于以标称轨迹 $\hat{e}$ 为中心、半径为 $\tilde{\rho}$ 的超管道内, 并且在所有时间内都成立, 即, $\|\rho(t)\| \leq \tilde{\rho}, \forall t \in \mathbb{R}_{\geq 0}$ 。

C. Online Optimal Control

如前所述, 方程 (23) 中的控制动作 $\hat{v}(\hat{e})$ 将是基于名义动力学方程 (22) 求解的 FHOCP 的结果。在这方面, 考虑一组采样时间 $\{t_k\}, k \in \mathbb{N}$, 采样周期为常数 $0 < \delta_t < T$, 其中 T 是预测时域, 使得 $t_{k+1} := t_k + \delta_t, \forall k \in \mathbb{N}$ 。在每个采样时间 t_k , FHOCP 的求解如下:

$$\min_{\hat{v}(\cdot)} \left\{ \|\hat{e}(t_k + T)\|_P^2 + \int_{t_k}^{t_k+T} [\|\hat{e}(s)\|_Q^2 + \|\hat{u}(s)\|_R^2] ds \right\} \quad (31a)$$

受以下条件约束:

$$\dot{\hat{e}}(s) = J(\hat{e}(s), p_d)\hat{v} + \zeta(\hat{e}(s), \dot{p}_d), \quad \hat{e}(t_k) = e(t_k), \quad (31b)$$

$$\hat{e}(s) \in \bar{\mathcal{E}}, \quad \hat{v}(s) \in \bar{V}, \quad \forall \delta_t \in [t_k, t_k + T], \quad (31c)$$

$$\hat{e}(t_k + T) \in \mathcal{F}, \quad (31d)$$

其中 $Q, P \in \mathbb{R}^{3 \times 3}$ 和 $R \in \mathbb{R}^{3 \times 3}$ 是正定增益矩阵。此外, $\mathcal{E}, V$ 和 $\mathcal{F}$ 是设计集合, 其定义是为了保证当 FHOCP (31a)-(31d) 的解是针对名义动力学 (22) 求得时, 实际轨迹 $e(t)$ 和控制输入 $v(t)$ 分别满足相应的状态和输入约束集合 $\mathcal{E}$ 和 V 。更具体地, 进行如下修改:

$$\bar{\mathcal{E}} := \mathcal{E} \ominus \mathcal{P}, \quad \bar{V} := V \ominus [-\sigma \circ \mathcal{P}]. \quad (32)$$

这直观地意味着集合 $\mathcal{E}, V$ 被相应收紧, 以保证在计算名义状态 $\hat{e}$ 和名义控制输入 $\hat{v}$ 时, 相应的实际误差状态 e 和实际控制输入 v 分别满足状态和输入约束 $\mathcal{E}, \mathcal{P}$ 和 $\mathcal{U}^2$ 。通过以下方式定义 *terminal set*:

$$\mathcal{F} := \{\hat{e} \in \bar{\mathcal{E}} : \|\hat{e}\|_P \leq \bar{\epsilon}\}, \quad \bar{\epsilon} > 0, \quad (33)$$

这里采用它是为了维持系统的稳定性[31]。

Newly Detected Obstacles: 如前所述, 工作空间内的障碍物可能通过车辆的车载传感器 (例如, 多波束成像或侧扫声纳) 在线检测。在这种情况下, 应确保 FHOCP 的解对应于车辆传感能力可到达的区域。这直观上意味着需要车辆在最坏情况下 (即机器人在最大干扰下的最大速度) 也能够看到任何潜在的新障碍物。因此, 假设 $\bar{R}$ 表示系统的感知范围, 如第 III-B 节中已经说明的那样, 预测视野 T 应设置如下:

$$T \leq \frac{\bar{R}}{\bar{V} + \tilde{\xi}} \quad (34)$$

其中 $\bar{V}$ 定义在 (13) 中。

备注3: 应注意, 在实际情景中, AUV 使用声纳传感器来获取有关

²This constitutes a standard constraints set modification technique adopted in tube-based NMPC frameworks. For more details see [34]).

environment. The detection range of these sonar sensors (i.e., $\tilde{R}$) depends on many factors, including the frequency. Low frequency sonars can detect objects at very long distance, depending on the sound propagation environment. Medium frequency sonars (typically operating between $7.5kHz$ and up to $30kHz$) can detect an object at a multiple nautical miles. On the other hand, high frequency sonars ($> 100kHz$), typically used for underwater inspection can detect smaller objects at a few hundreds meters (i.e., $> 100m$). On the other hand, as stated before, for needs of various common underwater tasks, the vehicle is required to move with relatively low speed. Thus, in view of (13), in a real scenario the predefined upper bound of the vehicle velocity can be tuned accordingly to the capability sensing range $\tilde{R}$ of the available sonar system in order to get a valuable prediction horizon enough for solving the FHOC (31a)-(31d).

Now we are ready to state the main result of this work:

Theorem 1: Suppose that at time $t = 0$ the FHOC (31a)-(31d) is feasible. Then, the proposed feedback control law (23), (27), renders the closed-loop system Input-to-State stable (ISS) with respect to the disturbances, for every initial condition $\hat{e}(0) \in \mathcal{E}$.

Proof: The proof of the theorem follows similar arguments presented in our previous work [35]. Due to the fact that only the state of the nominal system is used while the FHOC (31a)-(31d) is solved, the on-line optimization does not depend on the disturbances. Thus, the *feasibility proof* follows same arguments as in [31], [35], [36] and is omitted here. Regarding the *convergence analysis*, due to the fact that the set $\mathcal{P}$ is an RPI set, it holds that:

$$\|\rho(t)\| \leq \tilde{\rho}, \forall t \geq 0. \quad (35)$$

Due to the asymptotic stability of the nominal system, there exists a class $\mathcal{KL}$ function β such that:

$$\|\hat{e}(t)\| \leq \beta(\|\hat{e}(0)\|, t), \forall t \geq 0. \quad (36)$$

By combining (24), (35) and (36), we get:

$$\|e(t)\| \leq \beta(\|\hat{e}(0)\|, t) + \tilde{\rho},$$

for every $t \geq 0$, which leads to the conclusion of the proof. ■

V. SIMULATION RESULTS

In this section we consider a simulation study in order to demonstrate the efficiency of the proposed approach. The simulation results were conducted using a dynamic simulation environment built in MATLAB [37], [38] with sampling time $0.1sec$, which is common in a real time operation with an underwater robotic system. We considered the tracking control problem for an underactuated AUV operating in a workspace including two obstacles which lie in $\mathbf{p}_1 = [3, 0, 0]^T$ and $\mathbf{p}_2 = [-3, 0, 0]^T$ respectively. The desired trajectory is a circle in horizontal plane defined by $\mathbf{p}_d(t) = [3 \sin(\frac{\pi}{25}t), 3 \cos(\frac{\pi}{25}t), 0]^T$. Notice that the desired trajectory that is required to be tracked by the AUV coincides with obstacles positions. The predefined upper bound of the vehicle velocities are defined as: $\bar{u} = 0.4 \frac{m}{s}$, $\bar{w} = 0.3 \frac{m}{s}$

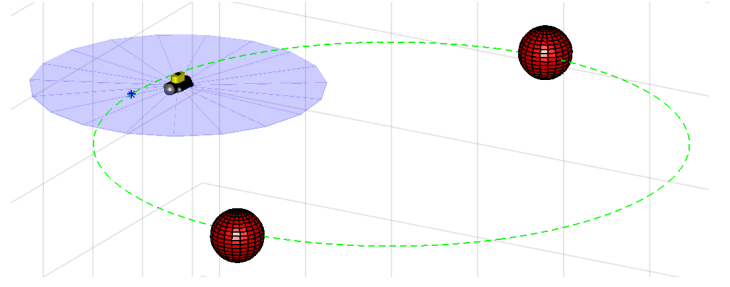


Fig. 4: Simulation setup: underactuated AUV operating in workspace including obstacles. The sensing range of the robot is indicated by a blue circle region around the robot. The desired trajectory is depicted with green dashed line which coincides with obstacles positions. The purpose of the controller is to track the real time evolution of the desired trajectory which is depicted by a blue star. The obstacles are detected and considered to the controller when they are within the sensing range $\tilde{R}$.

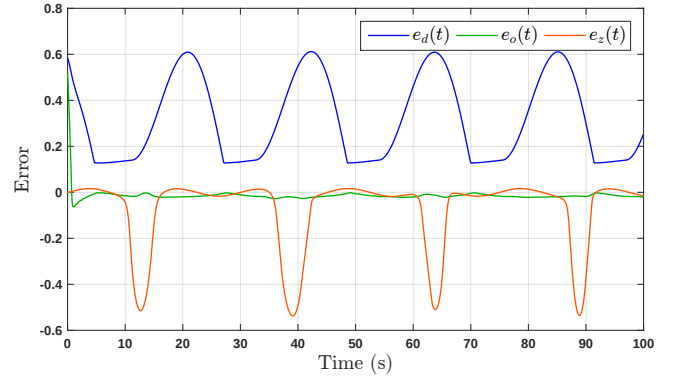


Fig. 5: The evolution of the error over time. The four peaks in e_d and e_z stand for the fact that the robot is leaving the desired trajectory in order to avoid the obstacles.

and $\bar{r} = 0.5 \frac{rad}{s}$. Moreover, the capability sensing range and the horizon of the FHOC are considered as $\tilde{R} = 1.5$ and $T = 8 * dt = 0.8sec$ respectively, satisfying the condition (34). Notice that the obstacles are detected and are considered by the controller when they are within the sensing range of the robot. The initial configuration is depicted in Fig 4. In addition, in the subsequent simulation study the dynamics of the considered AUV were affected by external disturbances in the form of slowly time varying sea currents acting along x , y and z axes modeled by the corresponding velocities $\omega_1 = 0.1 \sin(2\frac{\pi}{15}t) \frac{m}{s}$, $\omega_2 = 0.1 \cos(2\frac{\pi}{15}t) \frac{m}{s}$ and $\omega_3 = 0.1 \sin(2\frac{\pi}{15}t) \frac{m}{s}$. Furthermore, the parameter ϵ defined in (17) is set to $\epsilon = 0.1$. The simulation scenario has been conducted over a time period of 100 sec in which the robot is required to perform two encirclements, which means that it should avoid the obstacles twice each. The results are given in Fig. 5-Fig. 6. More specifically the errors evolution is depicted in Fig. 5. It should be noted that the four peaks in e_d and e_z stand for the fact that the robot is leaving the desired trajectory in order to avoid the obstacles. In addition, it can be observed that the error e_d remains always greater than $\epsilon = 0.1$, i.e., $e_d(t) \geq \epsilon, \forall t \geq 0$. Moreover, the control inputs evolution is depicted in Fig. 6. It can be witnessed that the control input constraints are always satisfied.

Video:

A video demonstrating the simulation of this section can be

环境。这些声纳传感器的探测范围（即 $\tilde{R}$ ）取决于许多因素，包括频率。低频声纳可以在非常远的距离探测物体，这取决于声波传播环境。中频声纳（通常工作在 7.5 kHz 到 30 kHz 之间）可以在多海里距离探测物体。另一方面，高频声纳（ $> 100\text{ kHz}$ ），通常用于水下检查，可以在几百米内探测较小物体（即 $> 100\text{ m}$ ）。另一方面，如前所述，为了满足各种常见水下任务的需求，车辆需要以相对较低的速度移动。因此，考虑到 (13)，在实际情景中，车辆速度的预设上限可以根据可用声纳系统的探测能力范围 $\tilde{R}$ 进行相应调整，以获得足够的预测视距，用于解决 FHOC (31a)-(31d)。

现在我们准备陈述这项工作的主要结果：

定理1: 假设在时间 $t = 0$ 时，FHOC (31a)-(31d) 是可行的。则所提出的反馈控制律 (23)、(27) 使闭环系统对于扰动输入-状态稳定 (ISS)，对于每一个初始条件 $\hat{e}(0) \in \mathcal{E}_0$ 。

Proof: 定理的证明遵循我们之前工作[35]中提出的类似论证。由于在求解 FHOC (31a)-(31d) 时仅使用标称系统的状态，因此在线优化不依赖于扰动。因此，feasibility proof 遵循与 [31]、[35]、[36] 相同的论证，在此省略。关于 convergence analysis，由于集合 $\mathcal{P}$ 是一个 RPI 集合，所以有：

$$\|\rho(t)\| \leq \tilde{\rho}, \forall t \geq 0. \quad (35)$$

由于标称系统的渐近稳定性，存在一个类别为 $\mathcal{KL}$ 的函数 β ，使得：

$$\|\hat{e}(t)\| \leq \beta(\|\hat{e}(0)\|, t), \forall t \geq 0. \quad (36)$$

通过结合 (24)、(35) 和 (36)，我们得到：

$$\|e(t)\| \leq \beta(\|\hat{e}(0)\|, t) + \tilde{\rho},$$

对于每个 $t \geq 0$ ，这得出了证明的结论。

五、仿真结果

在本节中，我们考虑一个仿真研究，以展示所提出方法的效率。仿真结果是在 MATLAB [37]、[38] 中构建的动态仿真环境下进行的，采样时间为 0.1 sec ，这在水下机器人系统的实时操作中是常见的。我们考虑了一个在工作空间中有两个障碍物的欠驱动 AUV 的跟踪控制问题，这两个障碍物分别位于 $p_1 = [3, 0, 0]^T$ 和 $p_2 = [-3, 0, 0]^T$ 。期望轨迹是在水平面上的圆，由 $p_d(t) = [3 \sin(\frac{\pi}{25}t), 3 \cos(\frac{\pi}{25}t), 0]^T$ 定义。注意，AUV 需要跟踪的期望轨迹与障碍物位置一致。车辆速度的预定义上限定义为： $\bar{u} = 0.4 \frac{m}{s}$ ， $\bar{w} = 0.3 \frac{m}{s}$

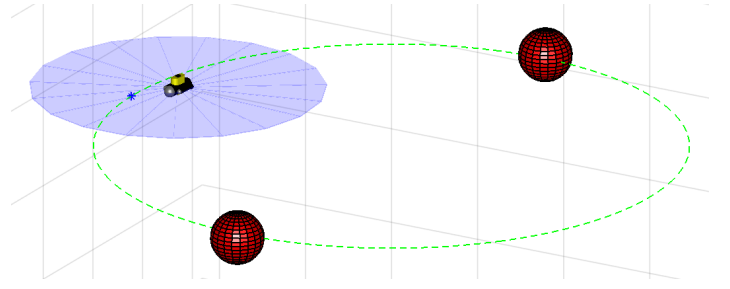


图4：仿真设置：欠驱动 AUV 在包含障碍物的工作空间中运行。机器人的感应范围由围绕机器人的蓝色圆区域表示。期望轨迹用绿色虚线表示，并与障碍物位置重合。控制器的目的是跟踪由蓝色星形表示的期望轨迹的实时演变。当障碍物在感应范围内时，会被检测并纳入控制器考虑。

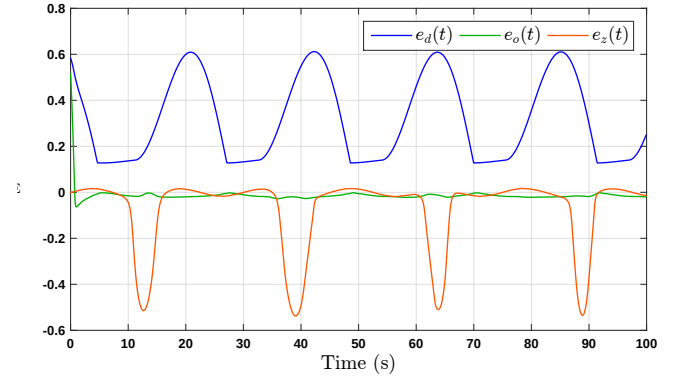


图5：随时间变化的误差演变。 e_d 和 e_z 中的四个峰值表示机器人正在偏离期望轨迹以避免障碍物。

且 $\bar{r} = 0.5 \frac{rad}{s}$ 。此外，FHOC 的能力感知范围和视野分别被认为是 $\bar{R} = 1.5$ 和 $T = 8 * dt = 0.8$ 秒，满足条件 (34)。注意，当障碍物位于机器人感知范围内时，会被检测并被控制器考虑。初始构型如图 4 所示。此外，在后续的仿真研究中，所考虑的 AUV 的动力学受到外部干扰的影响，这些干扰表现为沿 x 、 y 和 z 轴缓慢变化的海流，其对应的速度为 $\omega_1 = 0.1 \sin(2\frac{\pi}{15}t) \frac{m}{s}$ 、 $\omega_2 = 0.1 \cos(2\frac{\pi}{15}t) \frac{m}{s}$ 和 $\omega_3 = 0.1 \sin(2\frac{\pi}{15}t) \frac{m}{s}$ 。此外，在 (17) 中定义的参数 ϵ 被设定为 $\epsilon = 0.1$ 。仿真场景在 100 秒的时间周期内进行，其中机器人需要执行两次环绕，这意味着它应当每次避开障碍物两次。

视频：
可以演示本节模拟的视频

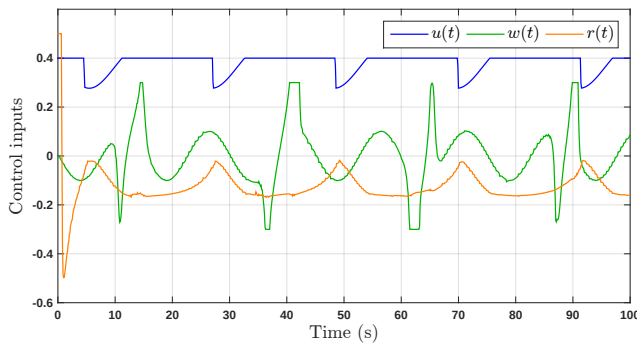


Fig. 6: The evolution of the control input signals over time.

found in the following link:

https://youtu.be/jJfjnp-mw_s

VI. CONCLUSIONS AND FUTURE WORK

This paper presents a robust trajectory tracking control for underactuated Autonomous Underwater Vehicles operating in a constrained workspace including obstacles. The purpose of the controller is to steer the underactuated AUV on a desired trajectory inside a constrained and dynamic workspace. The workspace knowledge (i.e., obstacles positions) is constantly updated online via the vehicle's sensors. Obstacle avoidance with any of the detected obstacles is guaranteed, despite the presence of external disturbances. Moreover, various constraints such as: obstacles, workspace boundaries, predefined upper bound of the vehicle velocity (requirements for various underwater tasks such as seabed inspection, mosaicking etc.) are considered during the control design. The proposed feedback control law consists of two parts: i) a Finite Horizon Optimal Control Problem (FHOC) and ii) a state feedback law which is tuned off-line and guarantees that the real trajectories remain inside a hyper-tube centered along the nominal trajectories. The closed-loop system has analytically guaranteed stability and convergence properties. Future research efforts will be devoted towards extending the proposed methodology for multiple Autonomous Underwater Vehicles operating in a dynamic environment including not only static but also moving obstacles.

REFERENCES

- [1] S. Heshmati-Alamdari, *Cooperative and Interaction Control for Underwater Robotic Vehicles*. PhD thesis, National Technical University of Athens, 2018.
- [2] S. Heshmati-Alamdari, C. Bechlioulis, G. Karras, and K. Kyriakopoulos, "Decentralized impedance control for cooperative manipulation of multiple underwater vehicle manipulator systems under lean communication," *IEEE OES Autonomous Underwater Vehicle Symposium*, 2018.
- [3] S. Heshmati-Alamdari, G. Karras, A. Eqtami, and K. Kyriakopoulos, "A robust self triggered image based visual servoing model predictive control scheme for small autonomous robots," *IEEE International Conference on Intelligent Robots and Systems*, vol. 2015-December, pp. 5492–5497, 2015.
- [4] T. Fossen, "Marine cybernetics," Norway, 2002.
- [5] C. Bechlioulis, G. Karras, S. Heshmati-Alamdari, and K. Kyriakopoulos, "Trajectory tracking with prescribed performance for underactuated underwater vehicles under model uncertainties and external disturbances," *IEEE Transactions on Control Systems Technology*, vol. 25, no. 2, pp. 429–440, 2017.
- [6] T. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. 2011.
- [7] J. H. Moon and H. J. Lee, "Decentralized observer-based output-feedback formation control of multiple unmanned underwater vehicles," *Journal of Electrical Engineering & Technology*, vol. 13, no. 1, pp. 493–500, 2018.
- [8] B. Subudhi, K. Mukherjee, and S. Ghosh, "A static output feedback control design for path following of autonomous underwater vehicle in vertical plane," *Ocean Engineering*, vol. 63, pp. 72–76, 2013.
- [9] A. Aguiar and A. Pascoal, "Dynamic positioning and way-point tracking of underactuated auvs in the presence of ocean currents," *Proceedings of the IEEE Conference on Decision and Control*, vol. 2, pp. 2105–2110, 2002.
- [10] A. Aguiar and J. Hespanha, "Position tracking of underactuated vehicles," *Proceedings of the American Control Conference*, vol. 3, pp. 1988–1993, 2003.
- [11] A. Aguiar and A. Pascoal, "Dynamic positioning and way-point tracking of underactuated auvs in the presence of ocean currents," *International Journal of Control*, vol. 80, no. 7, pp. 1092–1108, 2007.
- [12] F. Repoulas and E. Papadopoulos, "Planar trajectory planning and tracking control design for underactuated auvs," *Ocean Engineering*, vol. 34, no. 11-12, pp. 1650–1667, 2007.
- [13] J. Refsnes, A. Sørensen, and K. Pettersen, "Model-based output feedback control of slender-body underactuated auvs: Theory and experiments," *IEEE Transactions on Control Systems Technology*, vol. 16, no. 5, pp. 930–946, 2008.
- [14] D. Maalouf, A. Chemori, and V. Creuze, "L1 adaptive depth and pitch control of an underwater vehicle with real-time experiments," *Ocean Engineering*, vol. 98, pp. 66–77, 2015.
- [15] G. Antonelli, F. Caccavale, S. Chiaverini, and G. Fusco, "A novel adaptive control law for underwater vehicles," *IEEE Transactions on Control Systems Technology*, vol. 11, no. 2, pp. 221–232, 2003.
- [16] K. Do, J. Pan, and Z. Jiang, "Robust and adaptive path following for underactuated autonomous underwater vehicles," *Ocean Engineering*, vol. 31, no. 16, pp. 1967–1997, 2004.
- [17] L. Lapiere and B. Jouvencel, "Robust nonlinear path-following control of an auv," *IEEE Journal of Oceanic Engineering*, vol. 33, no. 2, pp. 89–102, 2008.
- [18] L. Lapiere, "Robust diving control of an auv," *Ocean Engineering*, vol. 36, no. 1, pp. 92–104, 2009.
- [19] S. Heshmati-Alamdari, G. Karras, and K. Kyriakopoulos, "A distributed predictive control approach for cooperative manipulation of multiple underwater vehicle manipulator systems," *IEEE International Conference on Robotics and Automation*, 2019.
- [20] B. Garau, M. Bonet, A. Alvarez, S. Ruiz, and A. Pascual, "Path planning for autonomous underwater vehicles in realistic oceanic current fields: Application to gliders in the western mediterranean sea," *Journal of Maritime Research*, vol. 6, no. 2, pp. 5–21, 2009.
- [21] C. Petres, Y. Pailhas, P. Patron, Y. Petillot, J. Evans, and D. Lane, "Path planning for autonomous underwater vehicles," *IEEE Transactions on Robotics*, vol. 23, no. 2, pp. 331–341, 2007.
- [22] Z. Zeng, L. Lian, K. Sammut, F. He, Y. Tang, and A. Lammas, "A survey on path planning for persistent autonomy of autonomous underwater vehicles," *Ocean Engineering*, pp. 303–313, 2015.
- [23] D. N. Subramani and P. F. Lermusiaux, "Energy-optimal path planning by stochastically dynamically orthogonal level-set optimization," *Ocean Modelling*, vol. 100, pp. 57–77, 2016.
- [24] H. K. Khalil, "Nonlinear Systems," Prentice Hall, 2002.
- [25] *The Society of Naval Architects and Marine Engineers, Nomenclature for treating the motion of a submerged body through a fluid (Technical and Research Bulletin No 105)*, 1950.
- [26] T. Fossen, "Guidance and control of ocean vehicles," Wiley, New York, 1994.
- [27] S. Heshmati-Alamdari, A. Eqtami, G. Karras, D. Dimarogonas, and K. Kyriakopoulos, "A self-triggered visual servoing model predictive control scheme for under-actuated underwater robotic vehicles," *Proceedings - IEEE International Conference on Robotics and Automation*, pp. 3826–3831, 2014.
- [28] D. Panagou and K. J. Kyriakopoulos, "Control of underactuated systems with viability constraints," *Proc. of the 50th IEEE Conference on Decision and Control and European Control Conference*, pp. 5497–5502, 2011.
- [29] D. Koditschek and E. Rimon, "Robot navigation functions on manifolds with boundary," *Advances in Applied Mathematics*, vol. 11, no. 4, pp. 412–442, 1990.
- [30] H. Michalska and D. Mayne, "Robust Receding Horizon Control of

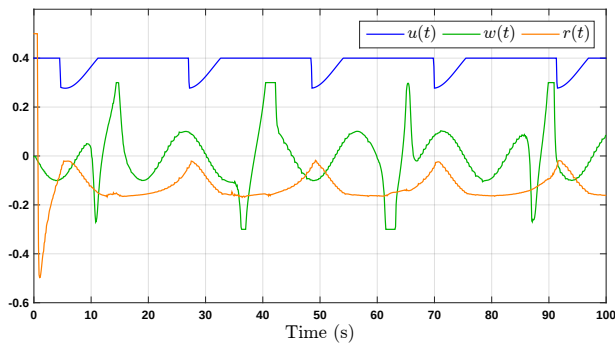


图6: 控制输入信号随时间的演变。

可在以下链接找到: <https://youtu.be/jJfjnp-mwS>

六、结论与未来工作

本文提出了一种用于在有障碍物约束工作空间中运行的欠驱动自主水下航行器 (AUV) 的鲁棒轨迹跟踪控制方法。该控制器的目的是将欠驱动AUV引导到受约束且动态变化的工作空间内的期望轨迹上。工作空间信息 (即障碍物位置) 通过车辆传感器不断进行在线更新。尽管存在外部干扰, 仍能保证与任何检测到的障碍物发生碰撞的避免。此外, 在控制设计过程中, 还考虑了各种约束条件, 例如: 障碍物、工作空间边界、车辆速度的预定义上限 (满足各种水下任务的需求, 如海底检测、拼接等)。所提出的反馈控制律由两部分组成: i) 有限时间最优控制问题 (FHOCP); ii) 离线调节的状态反馈律, 确保实际轨迹保持在超管状空间内。

参考文献

- [1] S. Heshmati-Alamdari, *Cooperative and Interaction Control for Underwater Robotic Vehicles*. 博士论文, 雅典国立理工大学, 2018.
- [2] S. Heshmati-Alamdari, C. Bechlioulis, G. Karras, 和 K. Kyriakopoulos, “多水下航行器操作系统协作操作的去中心化阻抗控制在低通信条件下,” *IEEE OES Autonomous Underwater Vehicle Symposium*, 2018.
- [3] S. Heshmati-Alamdari, G. Karras, A. Eqtami, 和 K. Kyriakopoulos, “一种稳健的自触发基于图像的视觉伺服模型预测控制方案用于小型自主机器人,” *IEEE International Conference on Intelligent Robots and Systems*, 卷 2015-12, 页 5492–5497, 2015.
- [4] T. Fossen, “海洋自动控制学,” *Norway*, 2002.
- [5] C. Bechlioulis, G. Karras, S. Heshmati-Alamdari, 和 K. Kyriakopoulos, “在模型不确定性和外部干扰下, 欠驱动水下航行器的具有规定性能的轨迹跟踪,” *IEEE Transactions on Control Systems Technology*, 卷 25, 第 2 期, 页 429–440, 2017.
- [6] T. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. 2011.

- [7] J. H. Moon 和 H. J. Lee, “基于分散观测器的多无人水下航行器输出反馈队形控制”, *Journal of Electrical Engineering & Technology*, 第 13 卷, 第 1 期, 页码 493–500, 2018 年.
- [8] B. Subudhi, K. Mukherjee 和 S. Ghosh, “自主水下航行器在垂直平面路径跟随的静态输出反馈控制设计”, *Ocean Engineering*, 第 63 卷, 页码 72–76, 2013 年.
- [9] A. Aguiar 和 A. Pascoal, “在洋流存在下欠驱动水下航行器的动态定位与航路点跟踪”, *Proceedings of the IEEE Conference on Decision and Control*, 第 2 卷, 页码 2105–2110, 2002 年.
- [10] A. Aguiar 和 J. Hespanha, “欠驱动航行器的位置跟踪”, *Proceedings of the American Control Conference*, 第 3 卷, 页码 1988–1993, 2003 年.
- [11] A. Aguiar 和 A. Pascoal, “在洋流存在下欠驱动水下航行器的动态定位与航路点跟踪”, *International Journal of Control*, 第 80 卷, 第 7 期, 页码 1092–1108, 2007 年.
- [12] F. Repoulas 和 E. Papadopoulos, “欠驱动水下航行器的平面轨迹规划与跟踪控制设计”, *Ocean Engineering*, 第 34 卷, 第 11-12 期, 页码 1650–1667, 2007 年.

- Constrained Nonlinear Systems,” *IEEE Transactions on Automatic Control (TAC)*, vol. 38, no. 11, pp. 1623–1633, 1993.
- [31] H. Chen and F. Allgöwer, “A Quasi-Infinite Horizon Nonlinear Model Predictive Control Scheme with Guaranteed Stability,” *Automatica*, vol. 34, no. 10, pp. 1205–1217, 1998.
 - [32] D. Mayne, J. Rawlings, C. Rao, and P. Scokaert, “Constrained Model Predictive Control: Stability and Optimality,” *Automatica*, vol. 36, no. 6, pp. 789–814, 2000.
 - [33] L. Grüne and J. Pannek, “Nonlinear model predictive control: Theory and algorithms,” *Communications and Control Engineering*, 2011.
 - [34] S. Yu, C. Maier, H. Chen, and F. Allgöwer, “Tube MPC Scheme Based on Robust Control Invariant Set with Application to Lipschitz Nonlinear Systems,” *Systems and Control Letters*, vol. 62, no. 2, pp. 194–200, 2013.
 - [35] A. Nikou and D. V. Dimarogonas, “Decentralized Tube-based Model Predictive Control of Uncertain Nonlinear Multi-Agent Systems,” *International Journal of Robust and Nonlinear Control (IJRNC)*, pp. 1–20, <https://doi.org/10.1002/rnc.4522>, 2019.
 - [36] A. Nikou and D. V. Dimarogonas, “Robust Tube-based Model Predictive Control for Timed-constrained Robot Navigation,” *American Control Conference (ACC)*, <https://arxiv.org/pdf/1809.09825.pdf>, Philadelphia, US, 2019.
 - [37] S. Heshmati-Alamdari, C. P. Bechlioulis, G. C. Karras, A. Nikou, D. V. Dimarogonas, and K. J. Kyriakopoulos, “A robust interaction control approach for underwater vehicle manipulator systems,” *Annual Reviews in Control*, vol. 46, pp. 315–325, 2018.
 - [38] S. Heshmati-Alamdari, A. Nikou, K. J. Kyriakopoulos, and D. V. Dimarogonas, “A Robust Force Control Approach for Underwater Vehicle Manipulator Systems,” *20th World Congress of the International Federation of Automatic Control (IFAC WC)*, Toulouse, France, vol. 50, Issue 1, pp. 11197–11202, 2017.

受约束非线性系统,” *IEEE Transactions on Automatic Control (TAC)*, 第38卷, 第11期, 第1623–1633页, 1993年。[31] H. Chen 和 F. Allgöwer, “一种具有稳定性保证的准无限时域非线性模型预测控制方案,” *Automatica*, 第34卷, 第10期, 第1205–1217页, 1998年。[32] D. Mayne, J. Rawlings, C. Rao 和 P. Scokaert, “受约束的模型预测控制: 稳定性与最优性,” *Automatica*, 第36卷, 第6期, 第789–814页, 2000年。[33] L. Grüne 和 J. Pannek, “非线性模型预测控制: 理论与算法,” *Communications and Control Engineering*, 2011年。[34] S. Yu, C. Maier, H. Chen 和 F. Allgöwer, “基于鲁棒控制不变集的管道MPC方案及其在利普希茨非线性系统中的应用,” *Systems and Control Letters*, 第62卷, 第2期, 第194–200页, 2013年。[35] A. Nikou 和 D. V. Dimarogonas, “不确定非线性多智能体系统的分散管道模型预测控制,” *International Journal of Robust and Nonlinear Control (IJRNC)*, 第1–20页, <https://doi.org/10.1002/rnc.4522>, 2019年。[36] A. Nikou 和 D. V. Dimarogonas, “用于定时的鲁棒管道模型预测控制